

Fourier interpolation of the q -space TDSCHA Lanczos without high-order force-constant tensors

TDSCHA development notes — `qspace_interpolation` branch

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Abstract

This note explains how to run the q -space TDSCHA Lanczos response on a fine uniform q mesh while the stochastic ensemble remains in a smaller coarse supercell. The physical motivation is simple: the response vector contains two-phonon blocks, and those blocks should sample a dense two-phonon continuum even when the expensive force calculations were done only in a coarse cell.

The central constraint is that we must not build high-order force-constant tensors. Instead, we interpolate the *per-configuration Bloch fields* that already enter the stochastic TDSCHA estimator. Windowing those fields induces a real-space image-assignment kernel on the averaged three- and four-field correlations. The report derives this kernel explicitly, shows why the plain off-grid Bloch sum is biased between commensurate points, and designs windows that reproduce the same minimal-image logic as centered tensor interpolation while keeping the runtime linear in the number of fine-mesh points.

Three invariants organize the construction. First, at commensurate wavevectors the interpolated operator must reduce exactly to the original coarse q -space Lanczos operator. Second, the interpolation kernel must respect the relevant permutation symmetries of the stochastic estimator. Third, acoustic sum rules must hold at the vertex level. We show how these conditions become linear constraints on the window kernel. For the third order this gives a tensor-free atom-resolved centering that reproduces the SnTe $2^3 \rightarrow 4^3$ Γ TO benchmark. For the fourth order the report documents why two simpler D4 centering attempts fail on SnTe and proposes a new four-leg, ASR-projected, factorized D4 kernel. The proposed D4 factorization is a low-rank representation of the *geometry kernel*, not of $\Phi^{(4)}$; the report proves its commensurate identity and its equivalence to the stochastic q -space D4 contraction. For the separate q -space fold/unfold formulation, a new atom-centred trigonometric cardinal kernel resolves the aliased Fourier images directly from the intracell coordinates. On SnTe it reconstructs the centred FC3 vertex norm to 1.2×10^{-5} relative accuracy and changes the stochastic $2^3 \rightarrow 4^3$ peak from the erroneous 46.11 to 38.04 cm^{-1} (centred oracle 37.65 cm^{-1}).

Contents

1 Notation and problem statement

This section fixes the notation used throughout the report and states the problem in operational terms.

symbol	meaning
N_c	number of cells, hence number of commensurate q points, in the coarse ensemble supercell
N_f	number of points in the fine interpolation mesh
n_{at}	number of atoms in the primitive cell
$n_b = 3n_{\text{at}}$	number of phonon bands at each q point
I	stochastic configuration index
$\mathbf{R}$	coarse-supercell lattice vector
$a, b, c, \dots$	atom and Cartesian indices, often collapsed into one index
$\mathbf{q}_p$	perturbing phonon wavevector of the response calculation
$\mathbf{q}_1, \mathbf{q}_2$	two-phonon wavevectors satisfying $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}$
$x_\nu(\mathbf{q}; I)$	mass-weighted displacement Bloch field in phonon mode ν
$y_\nu(\mathbf{q}; I)$	mass-weighted force-residual Bloch field in phonon mode ν
f_ψ	SSCHA displacement covariance factor $(1 + 2n)/2\omega$
f_Υ	inverse covariance factor $2\omega/(1 + 2n)$
$\Phi^{(n)}$	n -th order force-constant tensor, used only for comparison, not stored by the interpolation algorithm

The input data are the SSCHA ensemble configurations and forces in the coarse supercell, plus the converged SSCHA dynamical matrix. The desired output is the same q -space TDSCHA Lanczos operator that would be obtained from a direct fine-supercell ensemble, but evaluated from the coarse data. The important design restrictions are:

1. never construct or store $\Phi^{(3)}$ or $\Phi^{(4)}$;
2. keep setup and each Lanczos application linear in N_f ;
3. reproduce the original q -space operator exactly when the fine point is commensurate with the coarse mesh;
4. preserve Hermiticity, permutation symmetries, and acoustic sum rules.

The report is organized as follows. Sections ??–?? derive the stochastic interpolation kernel. Sections ??–?? add ASR and atom-resolved centering. Section ?? describes the implementation and cost. Sections ??–?? give the tests, including the current state of the fourth-order problem. Section ?? compares the resulting strategies with the interpolation schemes of the established anharmonic codes (phono3py, ShengBTE, D3Q/thermal2, and others) and states which one we recommend. The appendices repeat the two core derivations in a compact form.

2 Background: the q -space TDSCHA Lanczos

The q -space TDSCHA Lanczos [?, ?] computes a response function by repeatedly applying the linearized TDSCHA superoperator $\mathcal{L}$ to a vector with one one-phonon component and many two-phonon components. For a perturbation at $\mathbf{q}_p$ the vector has the schematic form

$$\psi = [R_\nu(\mathbf{q}_p) | a'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2) | b'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2)], \quad \mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}, \quad (1)$$

with one $n_b \times n_b$ block for each pair $(\mathbf{q}_1, \mathbf{q}_2)$ in the mesh. The harmonic part is simple once the SSCHA frequencies $\omega_\nu(\mathbf{q})$ and polarization vectors $e_\nu(\mathbf{q})$ are known. The only difficult part is the anharmonic action.

For each ensemble configuration I with statistical weight ρ_I , the code forms two Bloch fields:

$$\begin{aligned} x_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \sqrt{m_a} \tilde{u}_a(\mathbf{q}; I), & \tilde{u}_a(\mathbf{q}; I) &= \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}} u_a(\mathbf{R}; I), \\ y_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \frac{1}{\sqrt{m_a}} \tilde{f}_a(\mathbf{q}; I), & \delta f &= f - f_{\text{SSCHA}} - \langle f \rangle, \end{aligned} \quad (2)$$

where $\mathbf{R}$ runs over the N_c coarse-cell origins and the phase convention matches `vector_r2q`. The field x carries displacements; the field y carries force residuals. The Julia fused kernel contracts these fields into the TDSCHA action. Every anharmonic term is an ensemble average of either three fields (the D_3 part) or four fields (the D_4 part). For example, with $f_\Upsilon(\mathbf{q}, \nu) = 2\omega/(1+2n)$,

$$\left\langle \overline{(f_\Upsilon x)}(\mathbf{q}_p) x(\mathbf{q}_1) y(\mathbf{q}_2) \right\rangle, \quad \langle \bar{y}(\mathbf{q}_p) x(\mathbf{q}_1) x(\mathbf{q}_2) \rangle, \quad \dots \quad (3)$$

are representative D_3 averages, and the D_4 averages contain one more displacement field.

These averages are not heuristic. They are exact Gaussian integration-by-parts estimators of the anharmonic vertices. For instance, for the Gaussian SSCHA density with covariance Υ^{-1} ,

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial^3 V / \partial R_{a'} \partial R_{b'} \partial R_c \rangle, \quad (4)$$

up to the sign convention of the force. Disconnected Wick pieces vanish at SSCHA stationarity because $\langle u \rangle = 0$, $\langle \delta f \rangle = 0$, and $\langle u \otimes \delta f \rangle \propto \Upsilon^{-1} (\langle \partial^2 V \rangle - \Phi) = 0$. The same argument with one more displacement field gives the D_4 estimator. This is the key fact used throughout the report: the code estimates contractions of $\Phi^{(3)}$ and $\Phi^{(4)}$, but never needs to materialize those tensors.

3 Off-grid Bloch fields and the effective interpolation kernel

3.1 Why the naive off-grid transform is not an interpolation

The ensemble configuration is periodic with the coarse supercell. It therefore contains exact Fourier information only at the N_c commensurate wavevectors of that supercell. If one periodically repeats the configuration to infinity and Fourier transforms it at a genuinely incommensurate $\tilde{\mathbf{q}}$, the result is zero: the spectrum is a comb of delta functions at the commensurate points.

The code can nevertheless evaluate the finite sum in Eq. (??) at any $\tilde{\mathbf{q}}$. This is not wrong, but it has a precise meaning: it is a Fourier transform of one supercell multiplied by a rectangular window. The output is controlled by how this window assigns the periodic images of each real-space correlation to the phase factor $e^{-i\tilde{\mathbf{q}}\cdot\mathbf{R}}$. This is the same issue that tensor interpolation solves by centering: before Fourier transforming $\Phi^{(n)}$, one chooses the periodic image of every tensor element that has the shortest real-space geometry, and then zero-pads the rest.

The goal of stochastic centering is to reproduce that image assignment on the estimator itself. The fields are windowed before entering the stochastic averages; after ensemble averaging, those windows become an explicit real-space interpolation kernel.

3.2 The effective kernel of a windowed estimator

Consider first one D_3 product. The three slots are labelled (z, w, v) : the z slot carries the perturbation wavevector, and the w, v slots carry the two-phonon pair. Give each slot its own real window over the coarse-cell label $\mathbf{n}$:

$$\tilde{x}_W(\tilde{\mathbf{q}}; I) \propto \sum_{\mathbf{n}} W(\mathbf{n}) e^{-i\tilde{\mathbf{q}}\cdot\mathbf{R}_{\mathbf{n}}} u(\mathbf{R}_{\mathbf{n}}; I). \quad (5)$$

The window is zero outside its chosen support; the plain estimator is the special case $W \equiv 1$ on the whole coarse period. For a fine mesh that contains $\mathbf{q}_p$, the pair map enforces $\tilde{\mathbf{q}}_1 + \tilde{\mathbf{q}}_2 = \tilde{\mathbf{q}}_p + \mathbf{G}$

exactly. Translation invariance of the ensemble then gives

$$\langle \tilde{z}(\tilde{\mathbf{q}}_p) \tilde{x}_w(\tilde{\mathbf{q}}_1) \tilde{y}_v(\tilde{\mathbf{q}}_2) \rangle = \frac{1}{N_c^{3/2}} \sum_{\delta_1 \delta_2} S_{z,w,v}(\delta_1, \delta_2) c_3(\delta_1, \delta_2) e^{-i\tilde{\mathbf{q}}_1 \cdot \delta_1 - i\tilde{\mathbf{q}}_2 \cdot \delta_2}, \quad (6)$$

where c_3 is the physical three-point correlation at *actual* (unwrapped) lattice differences, the sum runs over all periodic images, and the **effective interpolation kernel** is the (per-dimension separable) window triple correlation

$$S_{z,w,v}(\delta_1, \delta_2) = \sum_s z(s) w(s + \delta_1) v(s + \delta_2). \quad (7)$$

Here δ_1 and δ_2 are actual, unwrapped lattice differences from the z slot to the w and v slots. The correlation $c_3(\delta_1, \delta_2)$ is the physical real-space three-point correlation. The function $S_{z,w,v}$ is the image-assignment kernel created by the windows. This equation is the main mathematical reduction of the report: interpolation quality is kernel quality.

Two consequences are immediate.

1. *Partition of unity.* For the plain window, $\sum_{\text{images}} S(\delta + L\mathbf{n}) = L$ for every difference class per dimension of length L . At commensurate $\tilde{\mathbf{q}}$, all images of the same folded class carry the same phase, so only this image sum matters. The estimator then reduces exactly to the coarse one.
2. *Tent bias off-grid.* The plain kernel assigns weight L – spread to each image: for $L = 4$, a nearest-neighbour correlation component leaks 25% of its weight onto an image three cells away, which enters with the wrong phase at incommensurate $\tilde{\mathbf{q}}$. The interpolation passes through the exact on-grid values but sags in between.

3.3 Equivalence with centered-tensor interpolation

Equation (??) states that the ensemble expectation of the off-grid estimator *is* the trigonometric (Fourier) interpolation of the real-space correlation function c_3 , with image weights S . Tensor interpolation performs the same Fourier sum after centering the tensor entries. Therefore:

If the stochastic kernel S equals the minimal-image indicator used by tensor centering, including tie splitting, then the expectation value of the windowed estimator is identical to the centered-tensor interpolation.

The difference is the order of operations. Tensor interpolation first forms a high-order real-space object and then contracts it. The stochastic method applies windows to the configurations and contracts the result configuration by configuration. The same centered object is obtained in the infinite-sampling limit, but the tensor is never stored. With the pre-filtering of Sec. ??, the interpolated correlation has the range of $\Phi^{(3)}$ itself, so this comparison is with the same short-ranged object used by `cellconstructor.Spectral`.

4 Stochastic centering: designed multitaper windows

4.1 Exactly solvable case: pair kernels

The previous section reduced the problem to kernel design. Before fitting the three-field kernel, it is useful to look at a two-field example where the answer is analytic. For a one-dimensional coarse period of length L , the minimal-image pair kernel is obtained exactly from two rectangular windows. With $r_m(\delta) = \max(0, m - |\delta|)$ the autocorrelation of $\text{rect}(m)$, for even L

$$\frac{1}{2} [r_{L/2+1}(\delta) - r_{L/2-1}(\delta)] = \begin{cases} 1 & |\delta| \leq L/2 - 1 \\ 1/2 & |\delta| = L/2 \quad (\text{Wigner-Seitz tie}) \\ 0 & \text{beyond,} \end{cases} \quad (8)$$

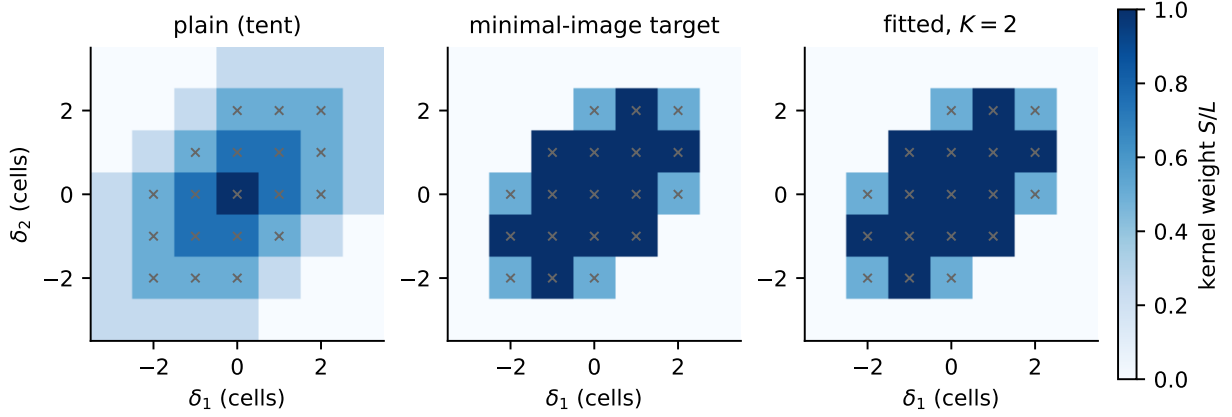


Figure 1: Effective interpolation kernels on the difference lattice (δ_1, δ_2) for a coarse dimension $L = 4$ (single hue: darker = larger weight; grey crosses mark the exact-image positions of the minimal-image target with their tie weights). Left: plain (tent) kernel of the naive off-grid transform. Middle: minimal-image target = zero-padded centering. Right: fitted $K = 2$ multitaper kernel — indistinguishable from the target (8×10^{-12} RMS).

including the correct tie splitting, and satisfying the partition of unity. This simple identity proves the principle: sharp minimal-image centering can be represented by separable windows applied to configurations, not only by explicit tensors.

4.2 Triple kernels: low-rank ALS design

For D_3 the target depends on two differences, because the three legs form a cluster. In one dimension define

$$\text{spread}(0, \delta_1, \delta_2) = \max(0, \delta_1, \delta_2) - \min(0, \delta_1, \delta_2).$$

The target $M(\delta_1, \delta_2)$ assigns weight to the image(s) with minimal spread and splits exact ties. This is the separable one-dimensional version of **ForceTensor**’s d_3 centering.

Using the same window in all three slots cannot improve on the tent kernel. The useful freedom is to allow the three slots to carry different windows. For each dimension we therefore fit K passes (z_r, w_r, v_r) by alternating least squares:

$$\sum_{r=1}^K \frac{1}{2} [S_{z_r, w_r, v_r} + S_{z_r, v_r, w_r}^T] \approx L M, \quad \text{subject to } \sum_{\text{images}} \sum_r S_r = L \text{ (hard partition constraint).} \quad (9)$$

The transpose average enforces the $w \leftrightarrow v$ symmetry of the two-phonon block. The hard partition constraint is what protects the exact commensurate limit. For the sizes used in the tests the fitted kernels are essentially exact ($L = 3, K = 3$: residual 2×10^{-16} ; $L = 4, K = 2$: 8×10^{-12}), as shown in Fig. ??.

In three dimensions the windows are products of one-dimensional factors, so K one-dimensional passes become K^3 three-dimensional passes. Each pass is just one call to the same Julia stochastic kernel. Optional origin averaging reduces variance; it must be implemented by rolling the configuration data while leaving the window support fixed. Shifting the window itself would wrap its support and would change the intended zero-padding.

5 Vertex renormalization $N_c \rightarrow N_f$

Changing the two-phonon mesh changes the normalization of mode-space vertices. A vertex written on an N -cell mesh scales as

$$d_3(\mathbf{q}_1\nu_1, \mathbf{q}_2\nu_2, \mathbf{q}_3\nu_3) = \frac{1}{\sqrt{N}} \delta_{\mathbf{q}_1+\mathbf{q}_2+\mathbf{q}_3, \mathbf{G}} \bar{\Phi}_3(\dots), \quad d_4 = \frac{1}{N} \delta_{\Sigma\mathbf{q}, \mathbf{G}} \bar{\Phi}_4, \quad (10)$$

where $\bar{\Phi}_n$ is intensive. The stochastic fields come from the coarse cell, so their raw averages carry the $N = N_c$ normalization. The interpolated Lanczos operator, however, is meant to act as the operator of an N_f -point fine mesh. Therefore the vertex factors must be converted:

$$s_3 = \sqrt{N_c/N_f}, \quad s_4 = N_c/N_f.$$

These are the `scale3` and `scale4` arguments of the Julia kernel.

A useful check is the one-loop bubble. The bubble has N_f pair terms, each containing $|d_3|^2$. With the scale factor, $|d_3|^2$ carries $s_3^2/N_c = 1/N_f$, so the total bubble is intensive and matches the direct fine-mesh normalization. The uniform per-order rescaling is exact up to disconnected Wick pieces $\propto \langle u \otimes \delta f \rangle$, which vanish at SSCHA stationarity; the ensemble should therefore be reweighted to the converged SSCHA dynamical matrix before interpolating. Empirically, *disabling* the rescaling makes the anharmonic renormalization overshoot by a factor $\simeq N_f/N_c$ (pinned by a regression test), confirming both its presence and magnitude.

6 Field pre-filtering: interpolating Φ -ranged objects

The windows should interpolate a short-ranged object. The raw stochastic correlations are not short-ranged enough, because each displacement leg carries the SSCHA covariance factor $f_\psi = (1 + 2n)/2\omega$. For example,

$$\langle x(\mathbf{q}_1) x(\mathbf{q}_2) y(\mathbf{q}_3) \rangle = f_\psi(\mathbf{q}_1) f_\psi(\mathbf{q}_2) [\Phi^{(3)} \text{ contraction}]. \quad (11)$$

In real space this correlation decays with the propagator range, including the acoustic $1/\omega$ tail, rather than with the range of $\Phi^{(3)}$. In the chain benchmark this caused a systematic underestimate of the anharmonic renormalization by about a factor two.

The fix is to move the covariance inverse $f_\Upsilon = 1/f_\psi$ from the Julia kernel into the coarse-grid fields before interpolation:

1. apply f_Υ exactly on the coarse grid, mode by mode, for each configuration;
2. transform the resulting filtered real-space configuration to the fine wavevectors;
3. set the kernel tables $f_\Upsilon, f_\psi \rightarrow 1$ (`prefiltered` flag);
4. fold the exact fine-side $f_\psi(\tilde{\mathbf{q}}_1) \otimes f_\psi(\tilde{\mathbf{q}}_2)$ into the $\alpha^{(1)}$ blocks (the α -contracted legs also use the filtered fields).

At commensurate points this is an exact rewriting of the old estimator, so backward compatibility is unchanged. Off-grid, the object being windowed now has the range of the force constants themselves. Frequencies, polarizations, χ , f_Υ and f_ψ are always evaluated from the interpolated SSCHA dynamical matrix on the fine mesh; no band-structure quantity is inferred stochastically.

7 Acoustic sum rules: a one-shot centering + ASR + symmetrization construction

Centering and the acoustic sum rule pull in different directions. A minimal-image kernel chooses compact images; an ASR-correct kernel must also vanish when any vertex leg is a uniform translation. In tensor codes this is why `ForceTensor.Center()` is followed by `Apply_ASR()`.

For the stochastic estimator the issue is even more concrete. Before windowing, the total force and the center-of-mass displacement constraints hold configuration by configuration. A nonuniform window cuts out part of the supercell, so those exact cancellations are no longer automatic. The problem is therefore to design a window kernel that:

1. has the desired centered image assignment off the coarse grid;
2. has exact partition sums, hence exact commensurate identity;
3. has zero coupling to an acoustic translation on any leg;
4. keeps the $w \leftrightarrow v$ permutation symmetry of the pair block.

The useful observation is that all these requirements are linear constraints on the kernel S .

7.1 What the ASR means at the kernel level

Let ϕ_{per} denote the coarse-periodic three-leg object being interpolated after pre-filtering. Translational invariance says that summing over any one physical leg gives zero. In difference coordinates and one lattice dimension of length L , this becomes the set of period sums

$$\sum_{d=0}^{L-1} \phi_{\text{per}}(d, \cdot) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(\cdot, d) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(d, d + \Delta) = 0 \quad \forall \Delta, \quad (12)$$

The third sum looks diagonal because the first leg is the reference leg of the difference coordinates. The interpolated object is the pointwise product $S \phi_{\text{per}}$ on the extended image grid. It satisfies the ASR for every possible ϕ_{per} obeying Eq. (??) if the image sums of S are class-independent:

$$\begin{aligned} (\text{ASR-}v) \quad & \sum_b S(\delta_1, d_2 + bL) \text{ independent of } d_2 \text{ for every fixed } \delta_1, \\ (\text{ASR-}w) \quad & \sum_a S(d_1 + aL, \delta_2) \text{ independent of } d_1 \text{ for every fixed } \delta_2, \\ (\text{ASR-}z) \quad & \sum_b S(d + bL, d + \Delta + bL) \text{ independent of } d \text{ for every fixed } \Delta. \end{aligned} \quad (13)$$

These are linear constraints on S . They are the kernel version of `Apply_ASR`. They also contain the partition condition: once the class sums are constant in each leg, only one scalar normalization is needed to recover the commensurate operator. Two checks follow.

- The plain (tent) kernel satisfies Eq. (??) exactly (its per-configuration sum rules are inherited untouched): the baseline estimator is ASR-clean everywhere.
- The *minimal-image kernel violates Eq. (??)* — the kernel-space restatement of “centering destroys the sum rule”, i.e. the very reason `Apply_ASR` exists. No admissible kernel that is exactly minimal-image can be ASR-exact at compact support.

7.2 A no-go theorem for one-period windows

If a window is supported on exactly one coarse period, its folded class sum is just the window value itself. Uniform class sums therefore imply the plain window. This proves a no-go statement: one-period nontrivial windows can improve centering, but they cannot also be ASR-exact at the kernel level.

A field-level projection is not a clean workaround. Subtracting a window-weighted acoustic zero mode modifies the effective window by a rank-one correction. That means the runtime kernel is no longer the kernel that was fitted. On the toy model this produced an $O(30\%)$ systematic bias. This is why the implementation applies such projections only for uniform or near-uniform windows.

7.3 The one-shot construction: doubled support, uniform class sums

The obstruction disappears if each window is allowed to live on two periods. The data are still periodic, but the Bloch phases are not wrapped: this is genuine zero-padding, just as in tensor centering. Parameterize a one-dimensional window by

$$w(d) = h(d), \quad w(d + L) = c_w - h(d), \quad d = 0 \dots L - 1, \quad (14)$$

with $h \in \mathbb{R}^L$ and c_w free. The folded class sum is $w(d) + w(d + L) = c_w$, independent of d . Thus uniform class sums are imposed by parameterization, not by a penalty, while the window can still be nontrivial.

On this manifold:

1. (ASR- v) and (ASR- w) hold identically; (ASR- z) is found to hold automatically for every fitted design (10^{-15} , no extra constraint needed).
2. Every per-pass kernel has *identical* class sums $L c_z c_w c_v$, so the full partition of unity (L^2 constraints) collapses to the single scalar equation $\sum_r c_z^r c_w^r c_v^r = 1$, enforced as a hard KKT row in each ALS slot fit: the commensurate limit is exact to machine precision. Indeed at commensurate $\tilde{q}$ the second-period Bloch phase is 1, each pass collapses to (class-sum product) $\times$ plain *configuration by configuration*, and the pass sum reproduces the plain estimator exactly (verified at 10^{-22}).
3. At $\tilde{q} = 0$ the effective weight is the constant class sum, so the acoustic contraction vanishes *per configuration with zero statistical noise* — the coarse-estimator mechanism restored. No field projection is needed or applied.
4. The $w \leftrightarrow v$ symmetrization is part of the kernel definition (orientation averaging), so it commutes with everything above.

This is the reason the stochastic kernel can be fitted in one shot. The centering target, ASR, partition, and symmetry are simultaneous linear conditions on the small kernel. In tensor space, by contrast, centering, ASR, and symmetrization are usually applied as separate projections.

The target cannot be the raw minimal-image kernel, because that kernel violates Eq. (??). The correct target is the projection of the minimal-image kernel onto the ASR and partition subspace. The projection distance is the unavoidable price of imposing exact ASR at finite support, analogous to the correction made by `Apply_ASR`:

RMS distance $LM \rightarrow$ projection	$L = 3$	$L = 4$	$L = 6$
support $2L$	0.144	0.248	0.374
support $3L$	0.072	0.127	0.193
plain (tent) kernel, for scale	0.315	0.462	—

The fitted windows reproduce the projected target nearly exactly (residual 0.028 at $L = 3$, $K = 3$; 0.071 at $L = 4$). Increasing support improves centering fidelity while preserving exact ASR. An optional metric weight ρ^{spread} can concentrate fidelity at short range, analogous to choosing how `Apply_ASR` spreads its correction. At runtime a doubled-support window on periodic data becomes the per- q complex weight $\hat{w}_{\tilde{q}}(s) = w(s) + w(s+L) e^{-2\pi i \tilde{q} \cdot L \mathbf{a}}$ per dimension. No change to the Julia kernels is required.

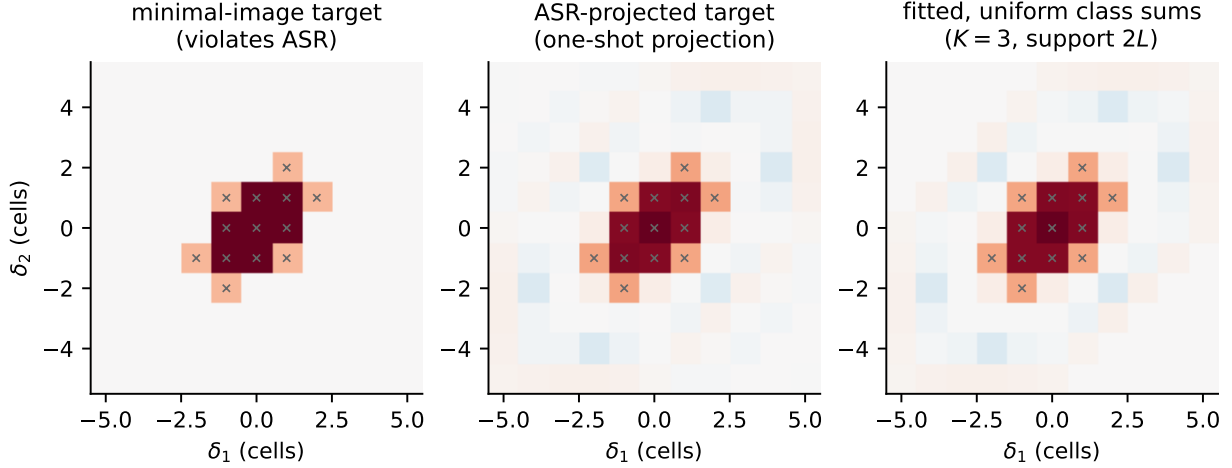


Figure 2: One-shot ASR-exact centering at $L = 3$ (diverging scale, red positive / blue negative; grey crosses = exact-image positions). Left: minimal-image target, which violates the kernel sum rules Eq. (??). Middle: its closed-form projection onto the ASR + partition subspace. Right: kernel realized by $K = 3$ doubled-support window passes with exactly uniform class sums — partition, all three sum rules and the $w \leftrightarrow v$ symmetry hold at 10^{-15} simultaneously.

7.4 Where the leak actually bites, and validation

Contracting each kernel with an exact ASR-satisfying test tensor gives the leak *deterministically* (Fig. ??, left): the translation contraction of the interpolated vertex on the acoustic leg $\tilde{q}_2 \rightarrow 0$ at fixed partner $\tilde{q}_1$. Two structural facts emerged that reshape where one should look for ASR artifacts:

- *The leak vanishes when both pair legs go to Γ together* ($q_p = 0$): the minimal-image violation carries phases that cancel at $\tilde{q}_1 \rightarrow 0$, and it re-vanishes at every commensurate $\tilde{q}_1$. The dangerous channel is an acoustic leg at fixed *finite* $\tilde{q}_1$ — physically, a finite- q phonon emitting a near- Γ acoustic phonon, exactly the process that builds the two-phonon continuum edge. There the minimal-image leak is $O(1)$ on the natural scale of the tensor (Fig. ??, left).
- *Pairwise potentials are structurally immune*. At $L = 3$ the violating image weights multiply only tensor entries whose three legs span three distinct cells; a two-body (bond) potential populates only two-site entries, so the contraction is identically zero. This is why the *M3* benchmarks (nearest-neighbour bond toy) never showed ASR artifacts — and why real crystals, whose $\Phi^{(3)}$ generically has three-site entries, will. The validation toy therefore adds a three-body term $g_{3b}s_1^2s_2$ spanning three cells.

The stochastic estimator confirms the picture quantitatively (Fig. ??, right; three-body-dominated chain, $L_c = 3$, fine mesh 48, perturbation at $q_p = 1/2$): as the acoustic partner $\tilde{q}_2 \rightarrow 0$ the plain and ASR-design vertices vanish linearly while the minimal-image one saturates at a plateau $\approx 7.6\times$ the plain value at the smallest probed $\tilde{q}_2 = 1/48$. The interpolated dynamical matrix keeps its own exact D2 ASR throughout (**Tensor2** centering + ASR), so the χ factors and the vertex vanish at the same points.

7.5 Response-level validation: leak versus resolution

The kernel-level ASR test is necessary but not sufficient; the real question is whether it affects the final response. In the three-body toy model, the static Lanczos observable $\omega_{\text{eff}} = \sqrt{-1/[M^{-1}]_{00}}$, three-body toy, $g_{3b}=0.08$, separates two independent errors.

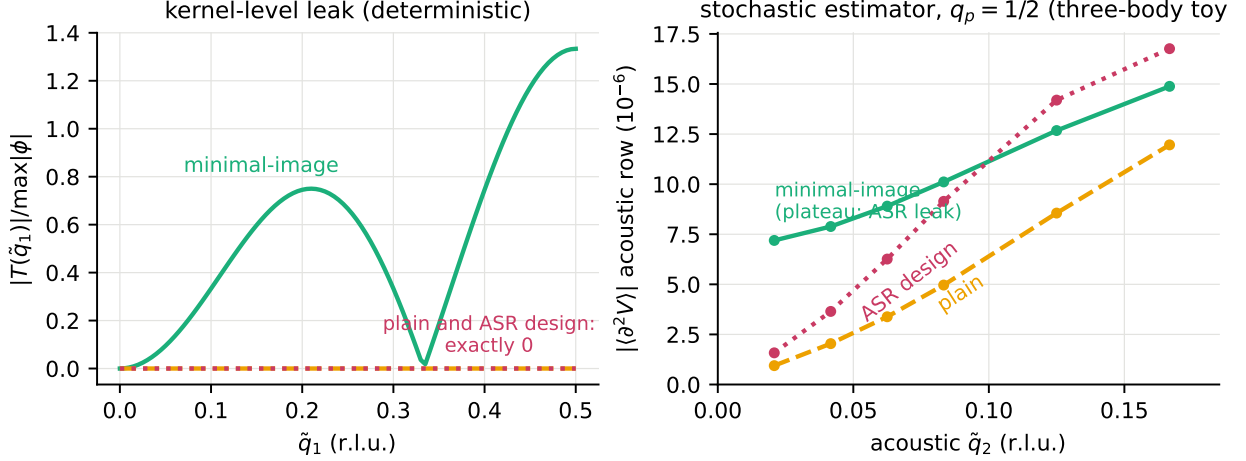


Figure 3: Left: deterministic kernel-level acoustic leak $|T(\tilde{q}_1)|$ (translation contraction of the interpolated three-body test tensor on the $\tilde{q}_2 \rightarrow 0$ leg, normalized to the largest tensor entry). The minimal-image kernel leaks $O(1)$ at generic $\tilde{q}_1$, vanishing only at Γ and at the commensurate point $1/3$; the plain and ASR-design kernels are exactly zero everywhere. Right: the same channel probed with the stochastic estimator ($N_I = 4000$, $q_p = 1/2$): the minimal-image acoustic vertex saturates (ASR leak) while plain and ASR designs vanish with $\tilde{q}_2$.

(i) *Under-resolved centering.* At $L_c = 3$, a spread-2 tensor sits exactly *on* the Wigner–Seitz tie boundary: the image assignment is Nyquist-ambiguous, and no compact-support kernel — centered, ASR-projected or otherwise — can resolve it. All oscillating designs then redistribute $O(1)$ tensor weight with wrong phases and fail catastrophically at strong coupling: aggregate renorm error 369.6 cm^{-1} for minimal-image (with 6 modes returning spurious non-negative-definite responses) and $\approx 300 \text{ cm}^{-1}$ for the ASR designs (8 such modes), against a two-seed noise floor of 48.1 and *plain* at 50.6 with no failures. This is a centering-resolution failure, not an ASR failure. It says that the coarse cell is too small for any centered interpolation. In that regime the plain tent kernel is the safe fallback.

(ii) *The ASR leak, at resolvable range.* At $L_c = 5$ (tensor interior to the WS cell) the roles invert — near- Γ interpolated modes, renormalization in cm^{-1} against a direct $L_f = 10$ ensemble:

mode	direct	plain	minimal-image	ASR design
$\tilde{q} = 1/10$, acoustic	−56	spurious $g > 0$	−77	−92
$\tilde{q} = 1/10$, optical	−28	−35	−31	−34
$\tilde{q} = 3/10$, acoustic	−413	−234	−396	−407
$\tilde{q} = 3/10$, optical	−793	−341	−733	−752

Here the *plain* kernel is the broken one: the tent bias halves the large renormalizations and flips a near-soft acoustic mode to a spurious non-negative-definite response, while minimal-image and ASR designs both track the reference, the ASR design closest on the strongly renormalized modes — with the qualitative difference that its vertex is exactly ASR-clean, whereas the minimal-image plateau of Fig. ?? is structural and gains weight as the fine mesh densifies the Γ neighbourhood.

The practical rule is therefore:

If the coarse cell resolves the third-order range, use the ASR design. If the range reaches the Wigner–Seitz boundary, no centered compact-support kernel is trustworthy and the plain estimator is the correct fallback.

On the original pairwise toy the ASR design keeps the same precision as the minimal-image windows (Sec. ??).

8 Atom-resolved pinned centering for $L = 2$ real-material cells

The cell-index windows above know only the supercell lattice. They do not see the basis offsets inside the primitive cell. This is fatal for the SnTe benchmark, whose coarse ensemble is only $2 \times 2 \times 2$. In such a cell every nonzero cell difference is a Wigner–Seitz tie at the cell-index level.

The standard tensor route nevertheless succeeds on the same data: `ForceTensor.Tensor3.Center(Far=3)` uses the actual Cartesian interatomic vectors, including the Sn and Te basis positions, and chooses the replica geometry with the smallest triplet perimeter. Thus the information is present in the coarse data; a tensor-free method simply has to use atom-resolved geometry instead of cell-only geometry.

The tensor-free fix is to pin one vertex leg to a primitive atom and coarse origin, then center the other legs relative to that actual atom. For one primitive atom a and one coarse origin $\mathbf{o}$, a perturbation-slot pin is

$$z_{a\mathbf{o}}(b, \mathbf{n}) = \delta_{ab} \delta_{\mathbf{n}\mathbf{o}}. \quad (15)$$

For a pair-leg atom b in cell-difference class δ , choose the periodic images that minimize the actual Cartesian distance

$$\mathcal{I}_{ab}(\delta) = \arg \min_{\mathbf{m} \in [-F, F]^3} |\tau_b - \tau_a + (\delta + \mathbf{m}\mathbf{L})\mathbf{A}|, \quad (16)$$

with exact ties split uniformly. The same atom-resolved rule is used for the two pair slots. Summing over all coarse origins restores the translation sum removed by the pinned delta; no $1/N_c$ prefactor is applied. The resulting D3 kernel is

$$K_{\text{at}}^{abc}(\delta_1, \delta_2) = \sum_{\mathbf{r}_1 \in \mathcal{I}_{ab}(\delta_1)} \sum_{\mathbf{r}_2 \in \mathcal{I}_{ac}(\delta_2)} \frac{e^{-i\tilde{\mathbf{q}}_1 \cdot \mathbf{r}_1} e^{-i\tilde{\mathbf{q}}_2 \cdot \mathbf{r}_2}}{|\mathcal{I}_{ab}(\delta_1)| |\mathcal{I}_{ac}(\delta_2)|}, \quad (17)$$

with $\mathbf{r} = (\delta + \mathbf{m}\mathbf{L})\mathbf{A}$. At runtime any selected image outside the fundamental supercell is folded back onto the periodic atom and multiplied by the corresponding Bloch phase $\exp[-i\tilde{\mathbf{q}} \cdot (\mathbf{m}\mathbf{L})\mathbf{A}]$; this is just the per- q , per-atom weight path already used by the doubled-support ASR windows. No $\Phi^{(3)}$ value enters Eq. (??).

The Julia D3 estimator is already permutation-symmetric, but that symmetry is expressed as three force-leg channels. One channel has the force on the perturbation slot z ; the other two have the force on w and v . The atom-resolved windows must follow the same split, otherwise they would center the wrong leg. The production path therefore runs three channel families:

$$\text{force on } z : \quad z\text{-pinned windows as in (??)}, \quad (18)$$

$$\text{force on } w : \quad w\text{-pinned windows, with } z, v \text{ centered relative to } w, \quad (19)$$

$$\text{force on } v : \quad v\text{-pinned windows, with } z, w \text{ centered relative to } v. \quad (20)$$

When all slot windows are identical, the sum of the three channel calls is algebraically identical to the original all-channel D3 kernel; this is a unit test. For SnTe the pass count is consequently $3 \times 2 \times 8 = 48$ D3 channel passes, but only 32 distinct field sets because many windows are reused.

The strict- Γ ASR follows from the same unit class sums. For every folded atom/cell class, the non-pinned windows have total selected-image weight one; after summing all pinned atoms and origins, the pinned slot has the same property. Thus a uniform acoustic displacement at $q = 0$ sees the same folded sums as the plain commensurate estimator and contracts to zero. Near Γ , however, the first moment of the selected images matters. That is why the final mode uses a conservative q -space acoustic projector (`atomic.delta`) instead of averaging away the pinned atom. A prototype with a uniform pinned slot lost the SnTe basis-offset information and reverted to the bad plain peak near 52 cm^{-1} .

Before using the construction in production, we performed an oracle test. The periodic SnTe $\Phi^{(3)}$ tensor is contracted with candidate geometry kernels, but only to measure the geometry

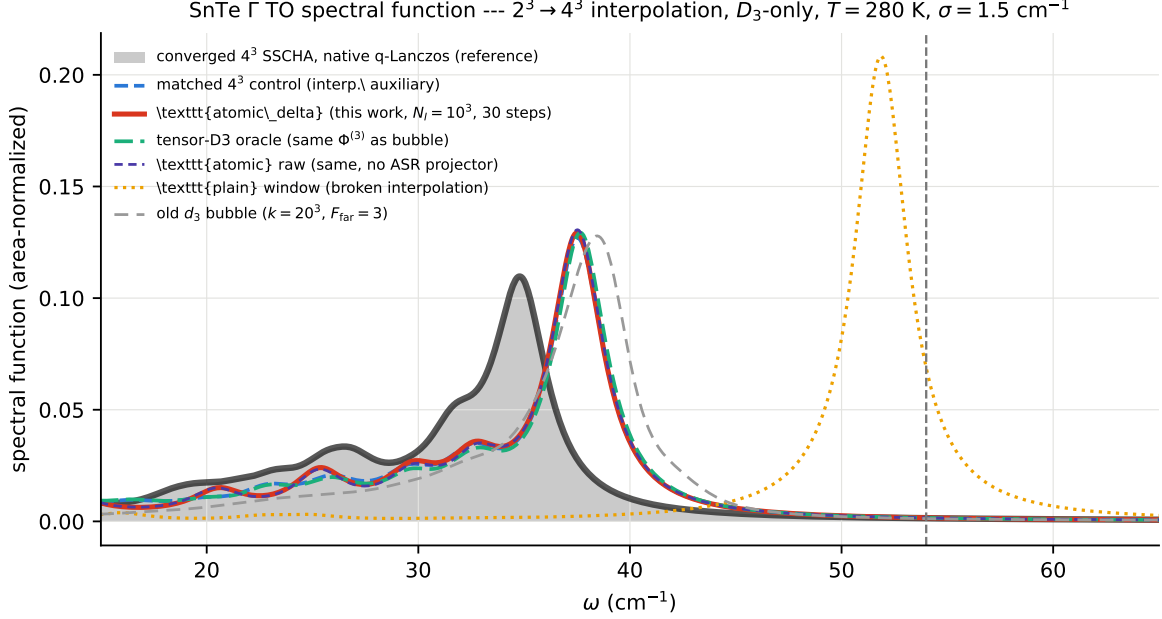


Figure 4: SnTe Γ TO spectral function, $2^3 \rightarrow 4^3$ interpolation at $T = 280$ K with 1.5 cm^{-1} smearing, D3-only (no D4). The converged 4^3 SSCHA reference (black shaded) is a native 4^3 q-Lanczos calculation from an independently converged 4^3 auxiliary dynamical matrix. The matched 4^3 control (blue dashed) used the same $2^3 \rightarrow 4^3$ interpolated auxiliary matrix as the window-valued methods and is retained only for algorithmic comparison. The `atomic_delta` curve (red) and tensor-D3 oracle (green) match the control; the `plain` window curve (yellow dotted) illustrates the broken $L = 2$ interpolation without atom-resolved centering.

error; the production estimator does not use this tensor. On the interpolated Γ -pair blocks of the failed $2^3 \rightarrow 4^3$ case, the plain cell-index kernel has mean relative Frobenius error 1.048 (max 1.582) against `Tensor3.Center(Far=3)`. The leg-wise atomic kernel has mean 0.008 (max 0.011), while the full perimeter target is zero within numerical precision. Thus the cheap leg-wise atomic kernel is already close enough to the validated tensor oracle.

The SnTe spectral-function benchmark confirms this. The interpolated Γ TO spectrum (D3-only, no LO-TO splitting) is shown in Fig. ?? and the peaks are summarised in Table ??.

The key result is that `atomic_delta` matches the matched 4^3 control (37.52 vs 37.56 cm^{-1}) and the tensor oracle (37.65 cm^{-1}), confirming that its atom-resolved window solves the cell-basis centering problem. The converged 4^3 SSCHA reference (34.79 cm^{-1}) is 2.73 cm^{-1} lower: this residual is not a centering error but the mesh convergence of the 2^3 auxiliary dynamical matrix, common to all methods that start from the same 2^3 SSCHA reference. The q -space acoustic projector dramatically improves the near- Γ acoustic sum rule. The acoustic-mode D_3 vertex norm at the smallest non-zero q on the 4^3 mesh ($|\mathbf{q}| \approx 0.066 \text{ \AA}^{-1}$) is reduced from 5×10^{-13} (raw atomic) to 3.4×10^{-17} under `atomic_delta` – a factor of ~ 14000 improvement (Fig. ??). At exact Γ both variants give machine zero because the folded-class-sum property of the permutation-correct atomic windows already enforces the hard partition.

The static response has $g > 0$ (unstable) for `atomic_delta`, `atomic`, and the tensor-D3 oracle, matching the matched 4^3 control; the broken `plain` interpolation instead produces a spurious stable response ($g < 0$, $w_{\text{eff}} = 49.5 \text{ cm}^{-1}$). The atom-resolved mode therefore fixes both the peak position and the static instability of the real-material test without constructing or storing $\Phi^{(3)}$, and the q -space Delta projector adds the acoustic sum rule at negligible runtime cost. Its current cost is the number of force-channel/pinned-atom/origin passes (SnTe: $3 \times 2 \times 8 = 48$ D3 pass calls). This makes `atomic_delta` the recommended tensor-free D3 mode for underresolved small supercells where cell-index windows cannot resolve atomic-basis ties.

Table 1: SnTe Γ TO peak positions and ASR quality at near- Γ wavevectors ($|\mathbf{q}| = 0.066 \text{ \AA}^{-1}$ on the 4^3 mesh). All Lanczos runs use $2^3 \rightarrow 4^3$ interpolation with $\sigma = 1.5 \text{ cm}^{-1}$; the tensor-D3 oracle and the converged 4^3 reference employ the deterministic d_3 tensor (centered, $F_{\text{far}} = 3$, ASR). N_I is the number of stochastic configurations; *n. s.* is the number of Lanczos steps. The acoustic D_3 row norm $\|D_3^{\text{ac}}\|$ is the Frobenius norm of the acoustic-band row of the d_{2v} blocks at $q_{\text{pert}} = \Gamma$; the ASR projector reduces this leakage by a factor of ~ 14000 without shifting the TO peak.

mode	peak (cm^{-1})	Δ vs conv. 4^3	N_I	n. s.	$\ D_3^{\text{ac}}\ $
converged 4^3 q-Lanczos	34.79	—	4000	250	—
matched 4^3 control (interp. aux.)	37.56	+2.77	4000	—	—
tensor-D3 oracle	37.65	+2.86	4000	250	—
<code>atomic_delta</code> (this work)	37.52	+2.73	1000	30	3.4×10^{-17}
<code>atomic</code> raw (same windows)	37.52	+2.73	1000	30	5.0×10^{-13}
<code>plain</code> window (broken)	51.83	+17.04	500	30	—
old d_3 bubble ($k=20^3$)	38.46	+3.67	—	—	—

8.1 Trilinear q-space interpolation and mesh convergence

A separate $2^3 \rightarrow 4^3$ interpolation strategy uses trilinear corner-summing of the coarse q -point vertex samples (`QSpaceTrilinearLanczos`). The independent test in Fig. ?? compares this approach against a converged 4^3 SSCHA reference (peak 34.79 cm^{-1} , independently relaxed auxiliary matrix). Standard atom-resolved centering brings the 2^3 spectrum from 44.03 down to 37.65 cm^{-1} ($+2.86 \text{ cm}^{-1}$ residual). Trilinear interpolation instead moves the peak *upward* to 46.11 cm^{-1} ($+11.32 \text{ cm}^{-1}$ error). The hardening is structural: trilinear corner-averaging cancels short-wavelength vertex components before the response squares them, attenuating the off-grid vertex to $\sim 29\%$ of its weighted norm. A decisive oracle test confirms this is the trilinear map itself, not a code bug: applying the same trilinear evaluation to the exact noise-free centered FC3 tensor reproduces the stochastic trilinear spectrum to within 0.13 cm^{-1} . Full details are documented in the benchmark accompanying this report.

8.2 Phase-cancellation audit at a cube centre

To isolate the cause of the trilinear amplitude loss, the vertex is evaluated at a single uncommensurate q point: the centre of a coarse-mesh cube, $q = (1/4, 1/4, 1/4)$ in fine-mesh fractional coordinates. The perturbation is at Γ and the response mode is the Γ TO (band 5). The same raw FC3 tensor (stored in the 2^3 supercell, *without* atom-triplet centering) is evaluated at each of the 8 coarse corners and at the fine centre via Fourier interpolation. The trilinear vertex is then the equal-weight ($1/8$) corner average of the coarse-corner samples; the centred reference comes from applying the tensor’s `Tensor3.Interpolate` directly at the fine q pair.

Table ?? lists the 8 corners and their contributions. The Frobenius norm of the projected D_3 vertex (TO-mode slice) is $\|\Phi_3\|_F = 1.47 \times 10^{-9}$ for the centred reference and 3.83×10^{-10} for the trilinear sum — a factor ~ 0.26 survival, consistent with the $\sim 29\%$ weighted $|V|^2$ ratio found in the full $2^3 \rightarrow 4^3$ benchmark.

The phases alternate because the Bloch factor $\exp(-2\pi i \mathbf{q} \cdot \boldsymbol{\tau})$ takes different values at each corner. In SnTe’s rocksalt cell, Sn sits at $(0, 0, 0)$ (phase $+1$ always), while Te at $(1/2, 1/2, 1/2)$ acquires the phases $+1, +i, -1, -i$ as the corner fractional coordinates toggle between 0 and $\pm 1/2$. The four corners with one non-zero index (*e.g.* $(0, 0, 1)$) have the largest individual $|\tilde{V}_{\text{TO}}| \sim 2 \times 10^{-10}$ but carry opposing phases ($\sim \pm 135^\circ, \sim \pm 90^\circ$); the equal-weight sum of their real parts nearly cancels, leaving a net amplitude two orders of magnitude smaller than the corner maxima.

Gauge transformations. Two candidate gauge transformations were tested at this q -point:

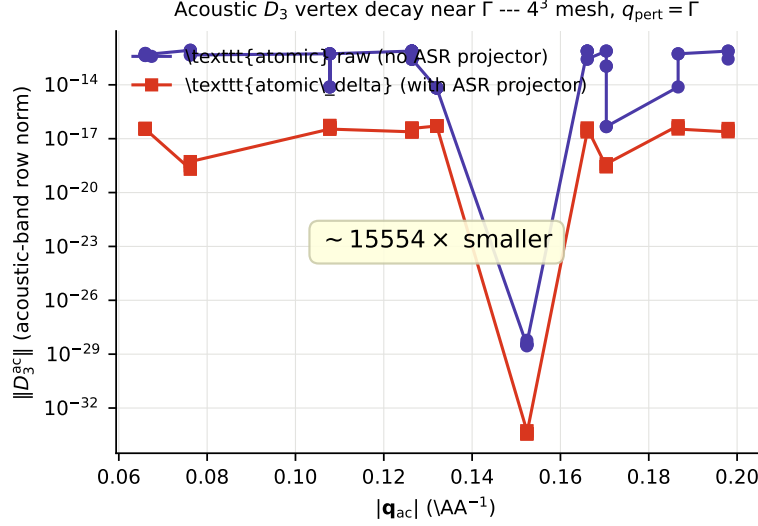


Figure 5: Acoustic D_3 vertex norm $\|D_3^{\text{ac}}\|$ (Frobenius norm of the acoustic-band row of the d_2v block at $q_{\text{pert}} = \Gamma$) versus the acoustic-leg wavevector $|\mathbf{q}_{\text{ac}}|$ on the 4^3 mesh. The `atomic_delta` projector (red squares) reduces the near- Γ acoustic leakage by about four orders of magnitude compared to the raw `atomic` mode (violet circles), while both modes give the same TO peak position (37.52 cm^{-1}).

- *Optimal scalar phase* per corner: align each corner's $\tilde{V}_{\text{TO}}$ phase to the centred reference before summing. This recovers only a factor $1.8\times$ more amplitude, far from the $1/0.26 \approx 3.8$ needed to match the centred norm. This is a hard ceiling for *any* phase-only correction: with positive weights, $|\sum_k w_k e^{i\varphi_k} V_k| \leq \sum_k w_k |V_k|$, and half of the corner samples here are essentially zero.
- *Atom-position Bloch gauge* $\exp(-2\pi i \mathbf{q} \cdot \boldsymbol{\tau}_a)$ as originally implemented: the Frobenius norm ratio drops from 0.26 to 0.17. This early test was *not conclusive*: the implementation evaluated the corner phases at BZ-wrapped representatives, which for half-integer $\boldsymbol{\tau}$ differences flips the sign of some corner phases (a branch inconsistency). Section ?? repeats the test with the branch-consistent transport and finds that the atomic gauge then aligns the phases essentially perfectly — but, per the previous bullet, still cannot restore the lost amplitude, and the D_3 -only Γ response turns out to be insensitive to vertex phases altogether.

Table 2: Eight coarse-cube corners contributing to the trilinear vertex at $q = (1/4, 1/4, 1/4)$ ($q_{\text{pert}} = \Gamma$, Γ TO band 5). The corner q indices are integers $\{0, 1\}$ on the 2^3 mesh; centred BZ fractions q_{cent} wrap $\pm 1/2$; weights are $1/8$. $\tilde{V}_{\text{TO}}$ is the TO diagonal element of the projected vertex; $\|\tilde{V}\|_F$ is the Frobenius norm of the full 6×6 D_3 block for the TO perturbation mode.

corner (k_1, k_2, k_3)	q_{cent} (frac)	weight	$ \tilde{V}_{\text{TO}} $	$\angle \tilde{V}_{\text{TO}}$	$\ \tilde{V}\ _F$	$ w \tilde{V}_{\text{TO}} $
(0, 0, 0)	(0, 0, 0)	1/8	6.8×10^{-19}	0.0°	1.5×10^{-9}	8.5×10^{-20}
(0, 0, 1)	(0, 0, $-1/2$)	1/8	6.2×10^{-14}	-62.8°	5.6×10^{-13}	7.8×10^{-15}
(0, 1, 0)	(0, $-1/2$, 0)	1/8	2.1×10^{-10}	$+135.3^\circ$	2.0×10^{-9}	2.6×10^{-11}
(0, 1, 1)	(0, $-1/2$, $-1/2$)	1/8	1.7×10^{-10}	-134.0°	2.3×10^{-9}	2.1×10^{-11}
(1, 0, 0)	($-1/2$, 0, 0)	1/8	2.0×10^{-10}	-135.8°	2.1×10^{-9}	2.5×10^{-11}
(1, 0, 1)	($-1/2$, 0, $-1/2$)	1/8	1.5×10^{-10}	$+90.0^\circ$	2.2×10^{-9}	1.9×10^{-11}
(1, 1, 0)	($-1/2$, $-1/2$, 0)	1/8	1.3×10^{-12}	$+89.4^\circ$	1.7×10^{-11}	1.6×10^{-13}
(1, 1, 1)	($-1/2$, $-1/2$, $-1/2$)	1/8	2.4×10^{-12}	$+85.0^\circ$	3.3×10^{-11}	3.0×10^{-13}
trilinear sum				5.2×10^{-11}		
centred ref. (Fourier)				1.9×10^{-14}		

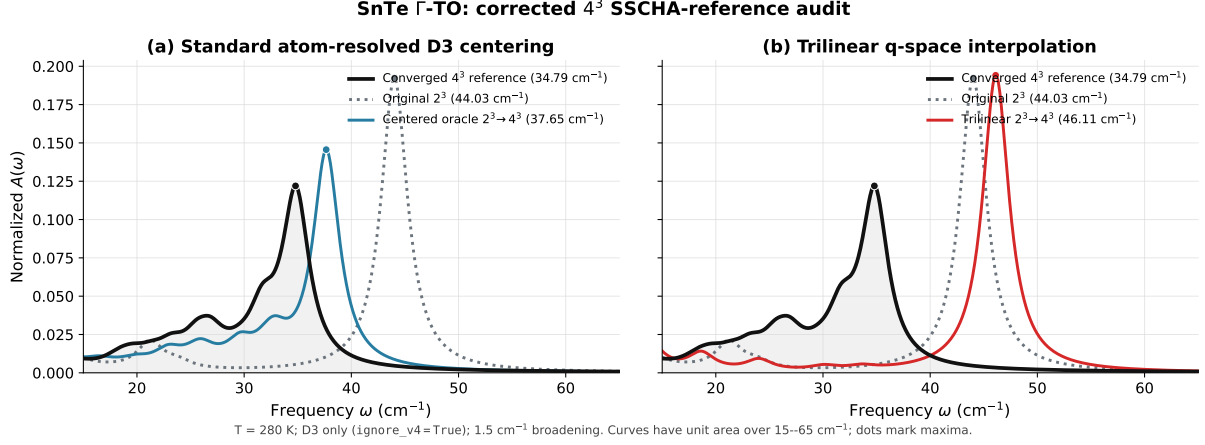


Figure 6: SnTe Γ -TO: corrected 4^3 SSCHA-reference audit. The converged 4^3 SSCHA reference (black) is a native 4^3 TDSCHA q-Lanczos calculation from an independently converged 4^3 auxiliary dynamical matrix (peak 34.79 cm^{-1}). Left panel (a): the standard atom-resolved centering oracle ($2^3 \rightarrow 4^3$) recovers the reference shape with a residual $+2.86 \text{ cm}^{-1}$ hardening. Right panel (b): trilinear q-space interpolation of the same coarse 2^3 data hardens the mode farther, to 46.11 cm^{-1} . Both methods start from the original 2^3 spectrum (grey dotted) at 44.03 cm^{-1} . $T = 280 \text{ K}$, D3 only, 1.5 cm^{-1} broadening. Curves have unit area over $15\text{--}65 \text{ cm}^{-1}$; dots mark maxima.

The information a phase cannot supply is the missing *amplitude* between the samples: the centred FC3 selects the minimal-distance triplet image, while the uncentred tensor uses a fixed cell-periodic image that couples Sn and Te images for $L = 2$, and the coarse samples sit at or near the nodes of the centred envelope. The route to precision-preserving interpolation therefore remains supplying the image assignment explicitly, either as atom-resolved FC3 centering (the tensor-oracle pipeline) or as the equivalent tensor-free centering windows of Sec. ???. Section ?? shows that this phase pattern is not an accident of the corner set but a quantized, inversion-protected property of the SnTe vertex, and Sec. ?? quantifies exactly how much of it a per-atom phase can and cannot repair. Section ?? then shows that the image assignment can also be supplied directly in q space by an atom-dependent trigonometric cardinal kernel, without constructing the tensor.

9 Why the phase of $\Phi^{(3)}$ rotates in q space: a topological view of the trilinear failure

This section steps back from the technical machinery and asks a simple question in simple language: *why* does the three-phonon vertex of SnTe acquire a complex phase that rotates as $\mathbf{q}$ moves through the Brillouin zone, why does the trilinear corner average consequently collapse to zero at the cube centres, and is there anything *quantized* — topological — behind this behaviour? The short answers, established numerically below (script `scripts/snte_phase_topology.py`, data `data/snte_phase_topology.{json,npz}`), are:

1. The phase rotation is real and remarkably clean: the dominant Te–Sn element of the vertex rotates *exactly linearly* by π (half a turn) every time $\mathbf{q}$ crosses the Brillouin zone once (Fig. ??c; the deviation from a straight line is 6×10^{-7} degrees).
2. The rotation happens because the anharmonic coupling is physically *centred on the Te–Sn bond*, i.e. at half a lattice vector from the points the interpolation grid can address. A half step in real space is a half turn of phase across the zone — the same shift theorem that says a delayed signal gets a linearly growing phase.

3. There is no divergence anywhere in the zone. The only points where the *true* vertex vanishes (and where its phase is therefore undefined) are symmetry zeros such as Γ . The extended “zeroing” happens only in the *interpolated* representation: the trilinear vertex is exactly real, and a real function connecting samples of opposite sign has no choice but to pass through zero.
4. The half-turn is quantized and protected by inversion symmetry: it is a $\mathbb{Z}_2$ invariant of the vertex, mathematically the same object as the Zak phase of the Su–Schrieffer–Heeger (SSH) chain. SnTe sits in the nontrivial class, and no small deformation of the force constants can move it out of that class.
5. The rotation itself *can* be stripped by a per-atom Bloch phase — provided the phase transport is branch-consistent (Sec. ??; an earlier implementation was not, which had hidden this). For SnTe the corrected atomic gauge is even the *optimal* phase, and it cuts the vertex reconstruction error from 112% to 18%. What no phase can restore is the *amplitude* lost between samples, and the D_3 -only Γ spectrum depends only on $|V|^2$: the corrected gauge moves the trilinear peak just $45.78 \rightarrow 45.59 \text{ cm}^{-1}$, out of an 8.1 cm^{-1} gap to the centred oracle.

9.1 What exactly is rotating

Fix the perturbation leg at Γ , project it on the TO mode, and look at the remaining object as a function of the internal wavevector,

$$V_{bc}(\mathbf{q}) = \sum_a e_a^{\text{TO}} \Phi_{abc}^{(3)}(\Gamma; \mathbf{q}, -\mathbf{q}), \quad (21)$$

a 6×6 matrix over the (atom, Cartesian) legs b and c . Split it into 3×3 atomic blocks. Two numerical facts organize everything:

- **The vertex lives on the Te–Sn bond.** The mixed Te–Sn blocks have norm ~ 0.28 (arbitrary units along the path), while the Te–Te and Sn–Sn blocks are three orders of magnitude smaller ($\sim 10^{-4}$). Whatever the interpolation does to the mixed blocks, it does to the whole self-energy.
- **The dominant element is a two-source interference.** Fourier-analysing the dominant Te–Sn element along the path $\mathbf{q} = (t, 0, 0)$ (reduced coordinates, $t : 0 \rightarrow 1$) gives exactly two lattice harmonics with *equal magnitude and opposite sign*, $c_{\mathbf{R}_1} = +0.2208$ at $\mathbf{R}_1 = (0, -2, 1)$ and $c_{\mathbf{R}_2} = -0.2208$ at $\mathbf{R}_2 = (1, -1, 0)$ (three-dimensional fit on an 8^3 q grid); everything else is below 10^{-3} of that. Hence, to an accuracy of 0.1%,

$$V(\mathbf{q}) = c \left(e^{2\pi i \mathbf{q} \cdot \mathbf{R}_1} - e^{2\pi i \mathbf{q} \cdot \mathbf{R}_2} \right) = \underbrace{2c \sin(\pi \mathbf{q} \cdot (\mathbf{R}_1 - \mathbf{R}_2))}_{\text{real envelope}} \underbrace{i e^{2\pi i \mathbf{q} \cdot \mathbf{x}_c}}_{\text{rotating phase}}, \quad (22)$$

with $\mathbf{x}_c = \frac{1}{2}(\mathbf{R}_1 + \mathbf{R}_2) = (\frac{1}{2}, -\frac{3}{2}, \frac{1}{2})$: the *midpoint between the two sources sits at a half-integer lattice position* — in fact exactly at the stored Te–Sn separation $\boldsymbol{\tau}_{\text{Te}} - \boldsymbol{\tau}_{\text{Sn}}$, i.e. on the Te–Sn bond.

Equation (??) is the whole story of Fig. ??. The modulus is a smooth sine arch that vanishes only where symmetry demands it (three Γ -point phonons of odd parity cannot couple in a centrosymmetric crystal, so $V(\Gamma) = 0$); the phase is the factor $e^{2\pi i \mathbf{q} \cdot \mathbf{x}_c}$, which advances by $2\pi \mathbf{G} \cdot \mathbf{x}_c = \pi \pmod{2\pi}$ — half a turn — across every zone crossing. In the complex plane the trajectory of $V(\mathbf{q})$ is a circle through the origin (Fig. ??a). The “rotation of the phase on the Te atoms” observed in the corner audit of Sec. ?? is exactly this factor: with Sn taken as the origin of the cell, the half-lattice-vector offset $\mathbf{x}_c$ is the position of the Te sublattice, so the rotation appears attached to the Te legs of the tensor.

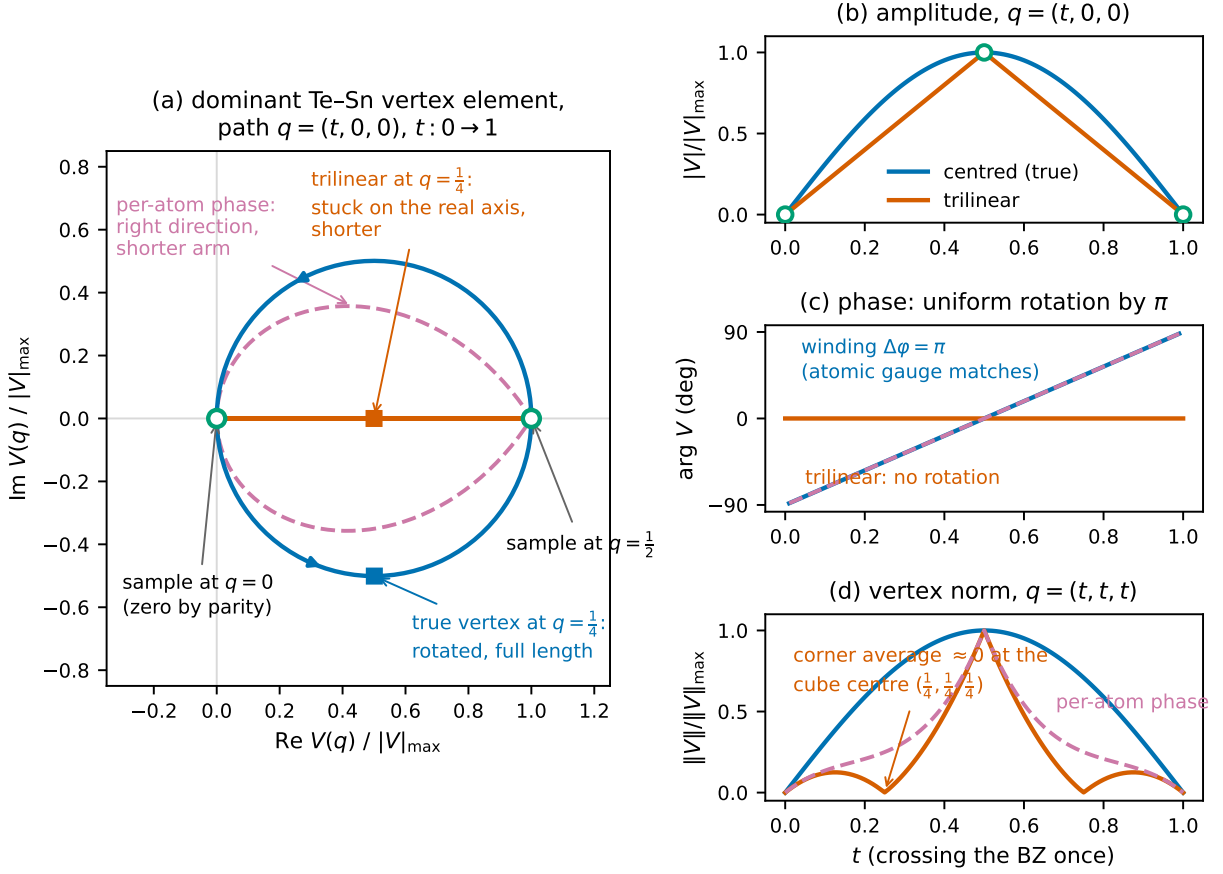


Figure 7: Measured phase winding of the SnTe three-phonon vertex (dominant Te-Sn element of Eq. (??), 2^3 force constants, centred Fourier continuation vs. trilinear corner reconstruction). (a) Complex-plane trajectory along $\mathbf{q} = (t, 0, 0)$, $t: 0 \rightarrow 1$: the true vertex (blue) travels on a circle through the origin — constant rotation of the phase — while the trilinear reconstruction (orange) is confined to the real axis, because all the coarse samples it can use (green circles) are real. At the off-grid point $q = \frac{1}{4}$ the true vertex is rotated by 45° and has full length; the trilinear one is real and shorter by $1/\sqrt{2}$. (b) Moduli along the same path: sine arch vs. tent. (c) Phases: the true phase advances linearly by exactly π per zone crossing (deviation from linearity 6×10^{-7} degrees); the trilinear phase cannot rotate at all. (d) Full TO-projected vertex norm along the cube diagonal $\mathbf{q} = (t, t, t)$: at the cube centre $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ the corner average cancels almost exactly ($\|V\|$ drops from 0.442 to 0.0007, a factor 1.6×10^{-3}), while the true norm is at 71% of its maximum there. Dashed curves: trilinear with the branch-consistent per-atom Bloch phase of Sec. ?? — the phase is now exact (panels a, c) and the cube-centre cancellation is lifted (panel d), but the amplitude stays damped between samples.

9.2 Why it rotates: the coupling sits between the sampling points

The rule behind Eq. (??) is elementary Fourier analysis: *a function whose weight sits at position $\mathbf{x}$ in real space acquires the phase $e^{2\pi i \mathbf{q} \cdot \mathbf{x}}$ in q space.* The interpolation grid addresses whole unit cells — integer lattice vectors. But in rocksalt SnTe the third-order coupling is not centred on a lattice site: it is centred on the Te–Sn nearest-neighbour bond, whose midpoint sits at half a lattice vector (Fig. ??a). Half a step in real space means half a turn of phase per Brillouin-zone crossing. The coarse $2 \times 2 \times 2$ q points are exactly the time-reversal-invariant points where this phase is 0 or π — so every coarse sample is *real* (largest imaginary part 10^{-12} of the real part, measured over all eight corners), with the sign pattern $(0, -, +, 0, +, 0, 0, -)$ over the eight corners. The samples are blind to the rotation; they only see its footprints, the alternating signs.

What the trilinear map then does is average real numbers with positive weights. Averaging two unit arrows 90° apart gives an arrow of length $1/\sqrt{2}$ (the chord of the circle instead of the arc, Fig. ??b) — this is precisely the measured $|V_{\text{tri}}|/|V_{\text{true}}| = 0.706$ at $\mathbf{q} = (\frac{1}{4}, 0, 0)$. Averaging two *opposite* arrows gives zero (Fig. ??c) — this is the cube-centre cancellation of Sec. ?? and Fig. ??d. Squaring these survival factors and multiplying over the three axes reproduces the order of magnitude of the 29–35% weighted $|V|^2$ survival measured in the full benchmark. The true vertex escapes through the complex plane; the interpolated one cannot, because no real convex combination of real samples ever leaves the real axis.

9.3 Is there a singularity in the Brillouin zone?

Not in the sense of a divergence — the vertex is a finite, smooth function everywhere. The singular object is the *phase*: it is undefined wherever $V(\mathbf{q}) = 0$, and a scan of the $(q_1, q_2, 0)$ plane shows that the only exact zeros of the true (centred) continuation are the parity-protected ones at Γ and its images; no interior phase vortices were found (every 2π plaquette winding on a 41×41 grid is zero). Around the Γ zero the trajectory of Fig. ??a passes through the origin, which is what lets a closed zone-crossing path accumulate a net phase without contradiction.

The zeros the user actually observes — the collapse “in the centre” — live in the *reconstruction*, not in the physics. The trilinear vertex is identically real (largest imaginary part 8×10^{-17} over the whole scanned plane, versus 0.44 for the true vertex), so its zero set is not a few isolated points but codimension-one surfaces separating regions of opposite sign, passing through the cube centres. In other words: the interpolation does not *find* a singularity of SnTe, it *manufactures* nodal surfaces because it lives on the real axis. One can also say it in gauge language: the phase factor $e^{2\pi i \mathbf{q} \cdot \mathbf{x}_c}$ with half-integer $\mathbf{x}_c$ is not periodic over the Brillouin zone, so there is no smooth *periodic* gauge in which the vertex becomes real everywhere. Removing the rotation requires importing the half-integer offset from outside the q samples — which is exactly what a branch-consistent per-atom Bloch phase does (Sec. ??). Any scheme that represents the vertex by real periodic data must pay with spurious zeros somewhere.

9.4 The quantized invariant: a Zak phase of the anharmonic vertex

The half turn is not an accident of this particular force field; it is pinned by symmetry. Write the dominant coupling channel as a set of lattice harmonics $c_{\mathbf{R}}$ as in Eq. (??). Because the crystal has inversion symmetry and $\Phi^{(3)}$ is odd under inversion (it is a third derivative), the coefficients must be *antisymmetric under reflection about the centre $\mathbf{x}_c$ of the coupling*:

$$c_{\mathbf{R}} = -c_{2\mathbf{x}_c - \mathbf{R}}, \quad 2\mathbf{x}_c \in \mathbb{Z}^3. \quad (23)$$

Consistency requires $2\mathbf{x}_c$ to be a lattice vector: the centre can sit *on a site* or *exactly half-way between sites*, nothing in between. This defines a $\mathbb{Z}_2$ -valued label per lattice direction,

$$\boldsymbol{\nu} = 2\mathbf{x}_c \bmod 2, \quad \text{SnTe: } \boldsymbol{\nu} = (1, 1, 1), \quad (24)$$

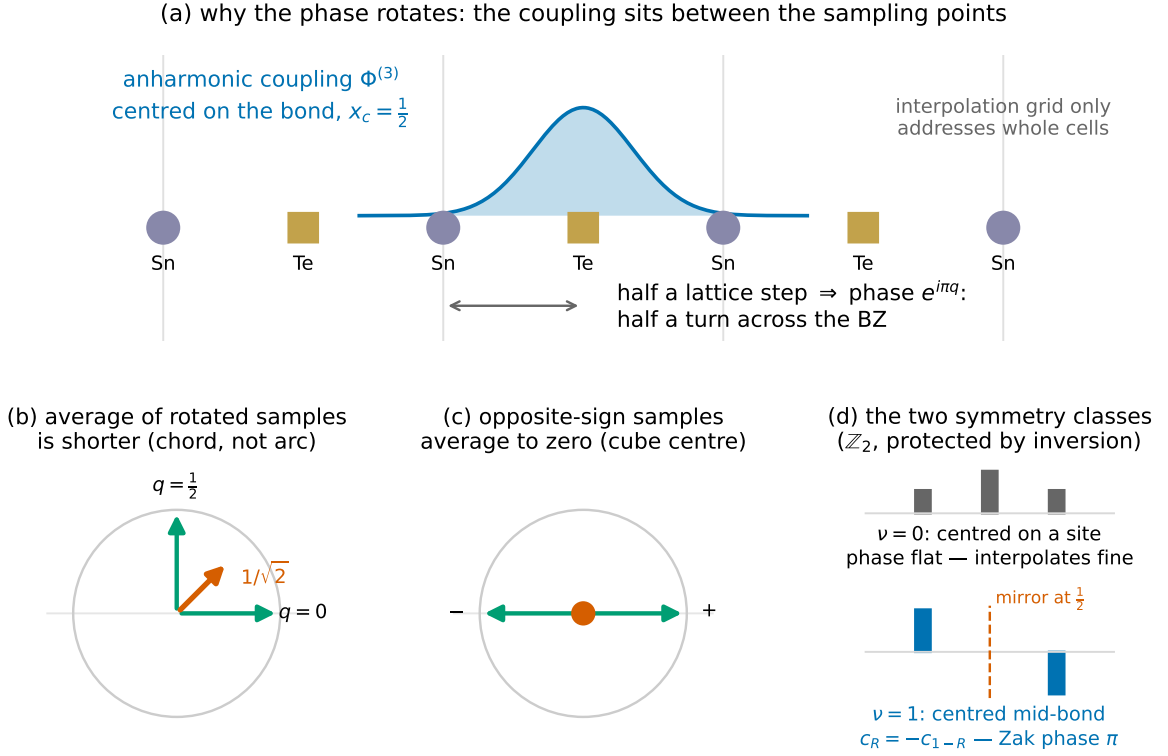


Figure 8: The topological picture in cartoons. (a) The three-phonon coupling of SnTe is centred on the Te–Sn bond, half a lattice step away from the cell origins that any cell-indexed interpolation can address; by the Fourier shift theorem its q -space phase therefore makes half a turn, $e^{i\pi q}$, across each Brillouin zone. (b) An interpolation with real positive weights replaces the arc of the rotating vertex by its chord: at a quarter zone the amplitude shrinks by $1/\sqrt{2}$. (c) Where the enclosed samples have opposite sign (cube centres), the chord passes through zero: the interpolated vertex vanishes although the true one is large. (d) Inversion symmetry allows only two classes for the centre of the coupling: on a lattice site ($\nu = 0$, Fourier coefficients mirror-symmetric about an integer) or exactly mid-bond ($\nu = 1$, coefficients antisymmetric about a half-integer, $c_R = -c_{1-R}$, Zak phase π). No continuous, symmetry-preserving change of the force constants connects the two classes. SnTe is in the $\nu = 1$ class along all three axes.

measured here from the fitted centre $\mathbf{x}_c = (\frac{1}{2}, -\frac{3}{2}, \frac{1}{2}) = \boldsymbol{\tau}_{\text{Te}} - \boldsymbol{\tau}_{\text{Sn}}$, i.e. from $c_{(0,-2,1)} = -c_{(1,-1,0)}$ satisfying Eq. (??) to one part in 10^3 (the same centre is found independently for all nine Cartesian elements of the Te–Sn block, see Sec. ??). The label ν is the direct analogue of the quantized Wannier-centre position (equivalently the Zak phase $\varphi_{\text{Zak}} = \pi\nu$) of the SSH chain: an inversion-symmetric 1D system can hold its charge centre either on the site or on the bond, the two classes are separated, and moving between them requires either breaking inversion symmetry or passing through a closing of the relevant amplitude. Fig. ??d sketches the two classes.

Three consequences follow for free, and all three are verified by the data:

- **Forced zero at Γ .** Summing Eq. (??) over $\mathbf{R}$ gives $\sum_{\mathbf{R}} c_{\mathbf{R}} = 0$ whenever $\nu \neq 0$: the dominant channel must vanish at Γ . (In SnTe this reinforces the independent parity selection rule for three zone-centre odd phonons.)
- **Real samples with alternating signs.** At time-reversal-invariant $\mathbf{q}$ the phase factor is ± 1 , so all coarse samples are real, and the sign alternates between the $q = 0$ and $q = \frac{1}{2}$ planes of each nontrivial direction — the pattern that makes the corner sums cancel.
- **Protection.** Because ν cannot change continuously, the rotation is invisible *inside* the coarse data: all samples are real, so nothing computed from the samples alone can recover the winding direction. The half-lattice-vector offset must be supplied from outside. The cheapest way to supply it is a branch-consistent per-atom Bloch phase, which repairs the phase of the reconstruction (Sec. ??) but cannot repair its amplitude; the complete cure remains the explicit image assignment (atom “centering”, i.e. the minimal-image windows of Sec. ??, or the centred-tensor evaluation), which restores the smooth real envelope as well.

For a general material the label is read off without any fit: compute $V(\mathbf{q})$ along one crossing of the zone per axis and record either the accumulated phase (an odd multiple of π in the nontrivial class) or simply the relative sign of the vertex at the time-reversal-invariant planes. Vertices with $\nu = 0$ (couplings centred on the sites of the interpolation lattice) are exactly the ones a corner-based scheme can interpolate accurately; every nontrivial direction contributes one factor of phase cancellation.

9.5 Can a per-atom phase cure it? Partly — here is exactly how much

A natural objection to the previous subsections runs as follows: if the whole problem is a phase $e^{2\pi i \mathbf{q} \cdot \mathbf{x}_c}$ that the interpolation cannot represent, why not *give* it that phase? Attach $e^{-2\pi i \mathbf{q} \cdot \boldsymbol{\tau}_a}$ to every atom, strip the phases before corner-averaging, and put them back afterwards. This objection is correct in an important way, and an earlier version of this report dismissed it with a wrong argument (“a unitary gauge does not change the singular values of the corner map”), which only constrains norms, not the alignment between the interpolated function and the truth. So we tested it properly (`scripts/snte_phase_gauge_test.py`, `scripts/oracle_gauge_spectrum.py`).

Two facts must be respected to make the test meaningful. First, the gauge transport must be *branch-consistent*: the phase carried from a fine point to its corners must be built from the local displacement $\boldsymbol{\delta} = \mathbf{x}_{\text{corner}} - \mathbf{q}$ (with the corner taken as the actual neighbour of $\mathbf{q}$, not wrapped back into the Brillouin zone). For half-integer $\boldsymbol{\tau}$ differences, independent BZ wrapping of corner and fine point shifts $\boldsymbol{\delta}$ by lattice vectors and flips signs — and this is precisely the bug in the previously implemented `atomic_phase` option, now fixed in `QSpaceTrilinear`. Second, for SnTe the per-atom phase turns out to be not merely helpful but *optimal*: fitting the antisymmetry centre of every Te–Sn coupling channel independently (all nine Cartesian elements) gives $\mathbf{x}_c = \boldsymbol{\tau}_{\text{Te}} - \boldsymbol{\tau}_{\text{Sn}}$ in every case, so the factorized atomic gauge coincides with the best possible per-channel phase to machine precision.

The measured outcome, over all 4^3 fine pairs (Γ TO row, relative errors in $\sum \|\cdot\|_F^2$):

reconstruction	rel. error ²	$ V ^2$ survival	$\ V\ $ at cube centre
plain trilinear	1.12	0.375	0.0007
atomic phase, wrapped (old, buggy)	0.78	0.422	0.156
atomic phase, branch-consistent	0.178	0.422	0.156
best integer branch of the gauge	0.178	0.422	0.156
per-channel oracle phase strip	0.178	0.422	0.156
truth	0	1	0.442

So the user-style objection is vindicated at the vertex level: the corrected per-atom phase eliminates the phase error almost completely (the residual 0.178 is pure amplitude damping), and the dashed curves in Fig. ?? show the reconstruction now rotating in step with the truth. But two ceilings remain, and they explain why the spectral benchmark barely moves:

- **Amplitude is phase-inaccessible.** With positive weights, $|\sum_k w_k e^{i\varphi_k} V_k| \leq \sum_k w_k |V_k|$ for any phases: a gauge can stop the cancellation but can never exceed the average of the sampled magnitudes. Half of the coarse samples sit at the nodes of the sine envelope (the parity zeros forced by $\nu \neq 0$), so at the cube centre even perfectly aligned samples recover at most half of the true amplitude; the measured survival saturates at 0.42.
- **The D_3 -only Γ response is phase-blind.** The two-phonon sector enters this observable as $|V(\mathbf{q})|^2 \chi(\mathbf{q})$ summed over pairs; vertex phases cancel identically. The deterministic oracle spectra confirm it: Fourier oracle 37.65 cm^{-1} , plain trilinear 45.78 , phase-corrected trilinear **45.59** — the optimal phase recovers only 0.19 of the 8.13 cm^{-1} gap, exactly the small $|V|^2$ gain in the table. This also retroactively explains the puzzling -0.13 cm^{-1} of the original (buggy) atomic-phase run: the bug corrupted phases, but this observable never looked at them.

The corrected gauge is still worth keeping (it is now the fixed behaviour of `atomic_phase=True`): it makes the interpolated *operator* right in phase, which matters for any quantity with vertex interference — $D_3 \times D_4$ cross terms, off-diagonal responses, perturbations away from Γ . The amplitude ceiling above is rigorous for *positive piecewise-linear weights*. It does not, however, prove that the missing envelope must be reconstructed by first building a real-space tensor. The next subsection tests the stronger version of the atom-centred Fourier proposal and changes that broader conclusion.

9.6 Atom-centred Fourier continuation: the centred result is recoverable in $\mathbf{q}$ space

The useful part of the proposed Fourier convention is not only a phase rotation. Define the Bloch displacement of atom a by

$$u_a^{\text{at}}(\mathbf{q}) = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} u_a(\mathbf{R}) e^{-2\pi i \mathbf{q} \cdot (\mathbf{R} + \boldsymbol{\tau}_a)}. \quad (25)$$

For the $\Gamma, \mathbf{q}, -\mathbf{q}$ vertex, a Cartesian element on atoms b, c then carries the relative offset $\mathbf{d}_{bc} = \boldsymbol{\tau}_b - \boldsymbol{\tau}_c$. Multiplying the cell-gauge vertex by $e^{-2\pi i \mathbf{q} \cdot \mathbf{d}_{bc}}$ removes the bond phase. Section ?? did exactly this, but then joined the two coarse values by a straight line. That last step throws away a second piece of information contained in Eq. (??): after the transformation the allowed Fourier harmonics are shifted by $\mathbf{d}_{bc}$ and the function need not be periodic at the boundary of the chosen cell.

A one-dimensional example makes the distinction concrete. The dominant SnTe element is, to excellent accuracy,

$$V(q) = A(e^{2\pi i q} - 1) = 2iAe^{i\pi q} \sin(\pi q). \quad (26)$$

The atom phase $e^{-i\pi q}$ makes it purely imaginary, but its amplitude is still the curved function $2|A|\sin(\pi q)$. From the two coarse samples $q = 0, 1/2$, linear interpolation gives only $|A|$ at $q = 1/4$, whereas the correct value is $\sqrt{2}|A|$. Thus the failure of phase-corrected trilinear interpolation is real. On the other hand, the atom-centred Fourier basis knows that the two shortest shifted harmonics are $-1/2$ and $+1/2$; their two-point cardinal interpolation reconstructs $\sin(\pi q)$ exactly. Restoring the atomic phase gives Eq. (??), including both its phase and amplitude.

We implemented this continuation without constructing $\Phi^{(3)}$. For each coarse Fourier alias $\mathbf{k}$ and atom pair b, c , choose the lattice image $\mathbf{R} \equiv \mathbf{k}$ (modulo the coarse mesh) that minimizes $|\mathbf{R} - \mathbf{d}_{bc}|$; exact distance ties are split. The Cartesian interpolation kernel is

$$P_{bc}(\mathbf{q}, \mathbf{n}) = \prod_{\alpha=1}^3 \frac{1}{N_{\alpha}} \sum_{k_{\alpha}=0}^{N_{\alpha}-1} \sum_{\substack{R_{\alpha} \equiv k_{\alpha} \pmod{N_{\alpha}} \\ R_{\alpha} \text{ nearest } d_{bc,\alpha}}} w_{R_{\alpha}} e^{2\pi i(q_{\alpha} - n_{\alpha}/N_{\alpha})R_{\alpha}}. \quad (27)$$

At every coarse $\mathbf{q}$ point this is exactly the identity. Away from the coarse mesh it is a complex, atom-pair-dependent cardinal function. The forward stochastic map uses $P^{\dagger}$ and the return map uses P , so the Lanczos operator remains Hermitian. Tests also verify $A(Q - k) = A(k)^T$ at both time-reversal-invariant and generic coarse Q . The implementation is selected by `QSpaceTrilinearLanczos(..., atom_fourier=True)`. The class name is historical: this option uses the same coarse-cost fold/unfold architecture, but Eq. (??) is trigonometric interpolation, not ordinary trilinear interpolation.

Correctness of the tie rule on non-orthogonal cells (audit, 2026-07-21). Equation (??) as written — and the original implementation — select the minimal image and split its ties *separably per Cartesian axis and in fractional coordinates*, i.e. metric-free: the product $\prod_{\alpha}$ chooses R_{α} nearest $d_{bc,\alpha}$ one axis at a time. This is exact only when the coarse-mesh Wigner–Seitz cell is an axis-aligned box, that is for *orthorhombic* coarse lattices with a diagonal metric. For any non-orthogonal cell the true minimal image minimizes the Cartesian length $|(\mathbf{R} - \mathbf{d}_{bc}) \mathbf{A}|$, which couples the three axes through the metric $\mathbf{A}\mathbf{A}^T$, and its degeneracies live on the non-axis-aligned faces of the Wigner–Seitz cell. The correct kernel is therefore the *non-separable* form

$$P_{bc}(\mathbf{q}, \mathbf{n}) = \frac{1}{N_1 N_2 N_3} \sum_{\mathbf{k}} \sum_{\substack{\mathbf{R} \equiv \mathbf{k} \pmod{\mathbf{N}} \\ \mathbf{R} \in \arg \min |(\mathbf{R} - \mathbf{d}_{bc}) \mathbf{A}|}} \frac{1}{\mu_{bc}(\mathbf{k})} e^{2\pi i(\mathbf{q} - \mathbf{n}/\mathbf{N}) \cdot \mathbf{R}}, \quad (28)$$

where the inner sum runs over the full set of metric-degenerate images of the aliasing class $\mathbf{k}$ and $\mu_{bc}(\mathbf{k})$ is its multiplicity. Equation (??) is the special case in which every such set factorizes into a Cartesian product of per-axis choices.

A direct geometry audit (`report/interpolation/scripts/audit_nyquist_ties.py`, `data/audit_nyquist`) makes the discrepancy concrete. On a cubic cell the two rules agree for every separation and every mesh tested, including the eight-fold Wigner–Seitz-corner tie, which the separable product reproduces correctly with weight $1/8$. On the *primitive fcc cell of SnTe* the two *kernels* disagree in the relative Frobenius norm by 23% (Te–Sn bond centre) to 36%; the separable rule can assign a class to a single image where the metric has a genuine four- or six-fold tie (the latter of weight $1/6$, a multiplicity no product of one-dimensional halves can represent), and it can also *manufacture* a spurious tie where the metric has a unique winner (e.g. fcc $\mathbf{d} = (\frac{1}{2}, 0, 0)$, class $(0, 0, 1)$: it splits $(0, 0, \pm 1)$ at $1/2$ while the metric selects $(0, 0, 1)$ alone).

Crucially, however, this kernel-norm discrepancy is *not* the error on a physical observable. The interpolation of a given coupling only ever excites the aliasing classes on which that coupling has non-zero coarse weight, and a short-ranged coupling excites only classes whose physically shortest image is *unique* — and on those classes the separable and metric rules agree. This is exactly the SnTe case: the dominant Γ -TO vertex lives on the two nearest-neighbour Te–Sn harmonics $\mathbf{R}_1 = (0, -2, 1)$ and $\mathbf{R}_2 = (1, -1, 0)$, each the *unique* shortest image of its class (3.28 Å, the

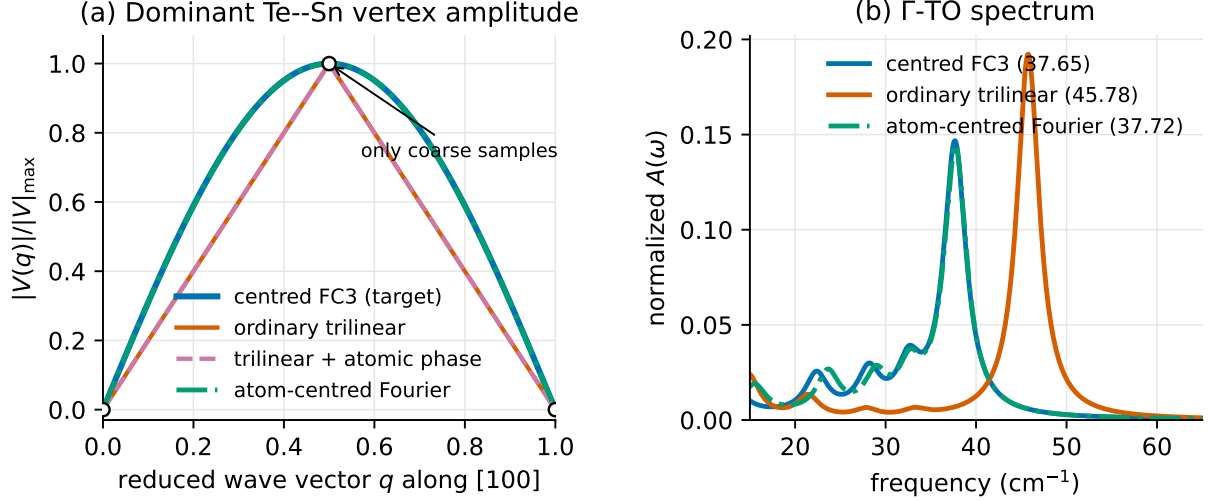


Figure 9: Why the atom position must change the interpolation basis, not only rotate phases. (a) The ordinary and phase-corrected trilinear maps join the two coarse amplitudes by straight segments. The atom-centred Fourier cardinal reconstructs the sine envelope and lies on top of the centred-FC3 target. Open circles are the only coarse samples used. (b) The corresponding deterministic tensor-D3 oracle spectra. The atom-centred q -space continuation recovers the centred spectrum, whereas ordinary trilinear interpolation hardens it by about 8.1 cm^{-1} .

next image at $7.34 \text{ \AA}$), so both rules place them identically. Reconstructing the actual two-image SnTe coupling from its 2^3 samples to the 4^3 mesh gives relative error² = 2.0×10^{-32} with the separable kernel and 2.2×10^{-32} with the metric kernel: *both machine-exact*. The 2.9×10^{-5} Γ -TO reconstruction reported above is therefore correct and is not compromised by the tie rule. The 23–36% kernel-norm figure lives entirely in classes the SnTe vertex does not excite (e.g. the $(0, 0, 0)$ class, whose coarse coefficient vanishes), and is multiplied by zero.

The defect bites only a coupling that carries weight in an aliasing class whose physically shortest image the separable rule mis-assigns — that is, a coupling sitting on a genuine metric degeneracy (a Wigner–Seitz face, edge or corner of a non-orthogonal cell) or on a separably-manufactured spurious tie. A minimal demonstration: on the primitive bcc cell with $\mathbf{d} = (\frac{1}{2}, \frac{1}{2}, 0)$, a coupling placed on the metric-shortest image of class $(0, 0, 1)$ — where the separable rule splits $(0, 0, \pm 1)$ at $1/2$ while the metric selects $(0, 0, -1)$ alone — is reconstructed with relative error² = 0.5 by the separable kernel and 1.1×10^{-32} by the metric kernel. Interestingly the user’s canonical bcc example, the central atom equidistant from its eight nearest neighbours ($\mathbf{d} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$), is *not* among the failing cases: its excited classes each reduce to a unique or separably-tied image, so the separable rule already handles it. The failure is thus not “every non-orthogonal cell always”, but specifically “couplings pinned to a non-axis-aligned Wigner–Seitz degeneracy”.

The scope is therefore: the one-dimensional chain benchmarks of Secs. ?? and ?? are unaffected (their $(1, 1, L)$ mesh has a single aliasing class along each transverse axis, a trivial factor of one, so the tie rule is never exercised), and the SnTe Γ -TO results are correct as reported; but for robustness on arbitrary non-orthogonal cells and couplings the metric-correct assignment of Eq. (??) is the implemented default from 2026-07-21 (`tie_metric=True`), with the separable form retained as an explicit orthorhombic fast path and as the control that exposes the discrepancy.

The numerical result is decisive. The first three rows below reconstruct the same centred Γ -TO FC3 row from the same 2^3 coarse samples; the spectra use the deterministic tensor only as a noise-free oracle for the interpolation map.

continuation	rel. error ²	$ V ^2$ ratio	$\ V\ $ at $(1/4)^3$	peak (cm ⁻¹)
ordinary trilinear	1.125	0.3750	0.00072	45.78
trilinear + branch-consistent atom phase	0.178	0.4219	0.1560	45.59
atom-centred Fourier, Eq. (??)	2.90×10^{-5}	0.999988	0.44131	37.72
centred Tensor3 Fourier oracle	0	1	0.44213	37.65

The production stochastic path gives 38.04 cm⁻¹ with only 1000 configurations and 30 Lanczos steps, consistent with the 37.65–37.72 noise-free controls. In other words, the unacceptable motion 44.03 → 46.11 cm⁻¹ of the original 2³ → 4³ trilinear calculation is removed; the new q-space result instead agrees with atom-resolved centering to within the stochastic resolution.

This finding resolves the apparent contradiction. It is indeed possible to recover the centred SnTe result using only q-space samples and atomic positions. What is impossible is to do so with the unique *piecewise-trilinear* function through those samples, even after a phase rotation. Equation (??) succeeds because it uses the atomic positions to resolve the aliased real-space image of each atom pair; it is the q-space form of pairwise atom centering. For the $\Gamma, \mathbf{q}, -\mathbf{q}$ SnTe slice this nearly equals full triplet centering. It is not yet a proof of equivalence for arbitrary external Q , for D4, or for a geometry where the best image depends essentially on all three atoms rather than on the two moving legs. Those cases remain tests of the more general atom-triplet kernel, not of the result established here.

9.7 What this says about SnTe — and what it does not

The invariant has a transparent physical reading: *the strong third-order anharmonicity of (this model of) SnTe is bond-centred*, living on the resonant Te–Sn nearest-neighbour bond rather than on either atom — consistent with the resonant-bonding picture commonly invoked for the giant anharmonicity of the IV–VI rocksalt compounds. Since the rocksalt structure puts that bond midpoint at half a lattice vector of the fcc cell, the nontrivial class $\boldsymbol{\nu} = (1, 1, 1)$ is a robust structural fingerprint: any IV–VI material with dominant nearest-neighbour anharmonic coupling will fail cell-cornered vertex interpolation in exactly the same way, while a material whose anharmonicity is site-centred (e.g. a rattler mode in a cage compound) would be in the trivial class and interpolate cleanly.

Two honest caveats. First, $\mathbf{x}_c$ is defined relative to the chosen cell origin; what is origin-independent is the *relative* statement that the coupling centre sits on the Te–Sn bond midpoint rather than on either sublattice — exactly like the Zak phase, which is origin-dependent while differences of Zak phases are physical. Second, this vertex invariant is *not* the electronic invariant that makes real SnTe a topological crystalline insulator; the two share the protecting symmetry (inversion/mirror on the rocksalt lattice) and the mathematical language, but they live on different objects (a three-phonon coupling function versus electronic Bloch bands). What they have in common is the lesson of this section: quantities pinned to half-integer positions by symmetry cannot be represented by smooth periodic data on the integer lattice, and any algorithm that pretends otherwise pays at the mid-points.

10 Implementation and cost

The implementation (branch `qspace_interpolation`) consists of `Modules/QSpaceInterpolation.py` and `tdscha.qspace.jl`. The Python class `QSpaceLanczosInterp` performs the fine-mesh setup: it builds the fine q list, hashes q points for $O(1)$ lookup, constructs the $O(N_f)$ pair map, interpolates the SSCHA dynamical matrix, builds the windowed Bloch fields, and dispatches the Lanczos action.

The alternative fold/unfold class `QSpaceTrilinearLanczos` is in `Modules/QSpaceTrilinear.py`. Its default path uses local piecewise-trilinear weights. The `atomic_phase=True` diagnostic keeps

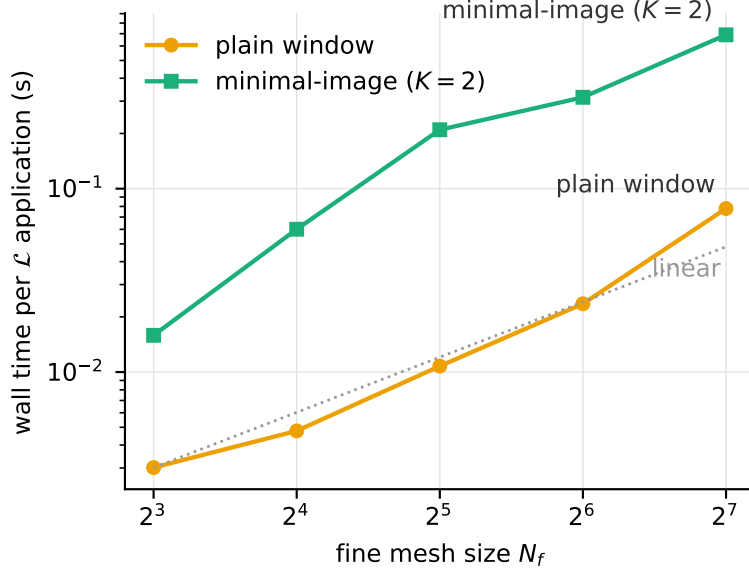


Figure 10: Wall time of one application of $\mathcal{L}$ versus the fine-mesh size N_f (chain toy, $N_I = 1000$ configurations, coarse $L_c = 4$): plain estimator and minimal-image windows ($K=2$, i.e. five kernel passes). Both follow the ideal linear trend (dotted).

those weights and transports the full-Bloch phase, while `atom_fourier=True` selects the atom-pair cardinal kernel of Eq. (??). The latter is cached at setup as $P(q, k, a, b)$; applying P or $P^\dagger$ costs $O(N_f N_c n_{\text{at}}^2)$ Cartesian scalar weights. Since N_c and n_{at} are fixed by the coarse calculation, the cost remains linear in the fine-mesh size.

The Julia side has two equivalent slot kernels. The scalar kernel is useful for readability and tests. The batched kernel groups configurations at a fixed symmetry operation: symmetry rotations become sparse-dense products, and the $\alpha^{(1)}$ intermediates and two-phonon accumulations become GEMMs. The scalar and batched kernels agree to 10^{-20} .

For a window design with R runtime passes, one application of the anharmonic part scales as

$$O(R N_{\text{conf}} n_{\text{sym}} N_f n_b^2).$$

For the separable minimal-image designs, $R = K^3$. For the plain estimator, $R = 1$. For the SnTe atom-resolved D3 mode of Sec. ??, $R = 3n_{\text{at}}N_c$ because there are three force channels, each pinned to one primitive atom and all coarse origins; in the 2^3 SnTe cell this is $3 \times 2 \times 8 = 48$ D3 passes. In all cases R is fixed by the coarse cell and the chosen kernel, not by the fine mesh, so the scaling in N_f is linear. Figure ?? verifies this over $N_f = 8 \dots 128$.

11 Benchmarks

The first validation problem is an anharmonic diatomic chain. It is small enough that we can generate both a coarse ensemble and a direct fine ensemble for the same model, so interpolation can be tested against a true fine-supercell reference. The model has harmonic springs K_1, K_2 and a bond cubic anharmonicity $g_3 s^3/3$; the weak-coupling tests use $g_3 = 0.1$, giving $5 \dots 40 \text{ cm}^{-1}$ renormalizations on $\sim 3000 \text{ cm}^{-1}$ modes. The cell is triclinic with symmetry $P1$, so point-group symmetries cannot hide sign, phase, or conjugation mistakes. Some test wavevectors are not time-reversal invariant ($\mathbf{q} \neq -\mathbf{q}$), which also checks the complex gauge conventions.

Exactness checks. These tests do not rely on physics judgment; they check identities that must hold exactly.

1. If the fine mesh equals the coarse mesh, the interpolated Lanczos coefficients reproduce the production `QSpaceLanczos` coefficients to 10^{-15} , both with and without pre-filtering.
2. The interpolated dynamical matrix reproduces the direct larger-supercell dispersion exactly for this range-1 harmonic model.
3. Rigid displacement shifts and constant force offsets leave the fields invariant.
4. The batched and scalar Julia kernels agree to machine precision, and the windowed path with trivial windows equals the plain path identically.

Physics: interpolation vs. direct fine supercell. The decisive comparison is between a coarse $(1,1,3)$ ensemble interpolated to the $(1,1,6)$ mesh and a direct $(1,1,6)$ ensemble of the same model. The error metric is the anharmonic renormalization $\omega_{\text{eff}} - \omega_{\text{SSCHA}}$, not the absolute frequency, because the harmonic part is already accurately interpolated and would otherwise dominate the comparison. Table ?? and Fig. ?? summarize the aggregate error $\sum |\Delta_{\text{int}} - \Delta_{\text{dir}}| / \sum |\Delta_{\text{dir}}|$:

	aggregate error	comment
noise floor (two direct seeds)	0.103	statistical reference
plain window	0.299	tent-kernel bias
minimal-image windows, 1 origin	0.158	
minimal-image windows, 3 origins	0.122	at the noise floor
ASR design, 3 origins (Sec. ??)	0.150	exact ASR, no precision lost
ASR design + decay weight $\rho=0.4$	0.196	
plain, <i>no</i> pre-filter	$\approx 2\times$ deficit	systematic (Sec. ??)
plain, <i>no</i> s_3, s_4 rescaling	$\approx 2\times$ overshoot	systematic (Sec. ??)

Table 3: Aggregate renormalization error of the interpolated Lanczos against direct fine-supercell calculations (chain toy, $L_c = 3 \rightarrow L_f = 6$, $N_f = 4000$, all 33 valid modes). The designed windows remove essentially all resolvable interpolation error; the last two rows document the failure modes that the pre-filter and the vertex rescaling correct.

12 Example application: spectral functions at interpolated q and the standard d_3 bubble

As an end-to-end check, we compute one-phonon spectral functions at wavevectors absent from the coarse mesh and compare them with the standard d_3 dynamic bubble from `cellconstructor.Spectral.get_dia`. In this toy model the analytic $\Phi^{(3)}$ is known, so the bubble route can use the exact centered and ASR-projected tensor. The Lanczos calculation disables D_4 so that both methods contain the same diagrammatic content. Because the model is weakly coupled, the perturbative bubble and the resummed TDSCHA Lanczos should agree. They do: Fig. ?? shows agreement at the energy-grid resolution, about 2.3 cm^{-1} , on shifts of $15 \dots 40 \text{ cm}^{-1}$.

q	band	ω_{SSCHA} (cm^{-1})	peak (Lanczos interp) (cm^{-1})	peak (d_3 bubble) (cm^{-1})
2/16	4	2205.3	2185.6	2183.2
2/16	5	2250.6	2214.2	2216.6
6/16	4	2013.4	1994.9	1992.5
6/16	5	2058.5	2040.2	2040.2

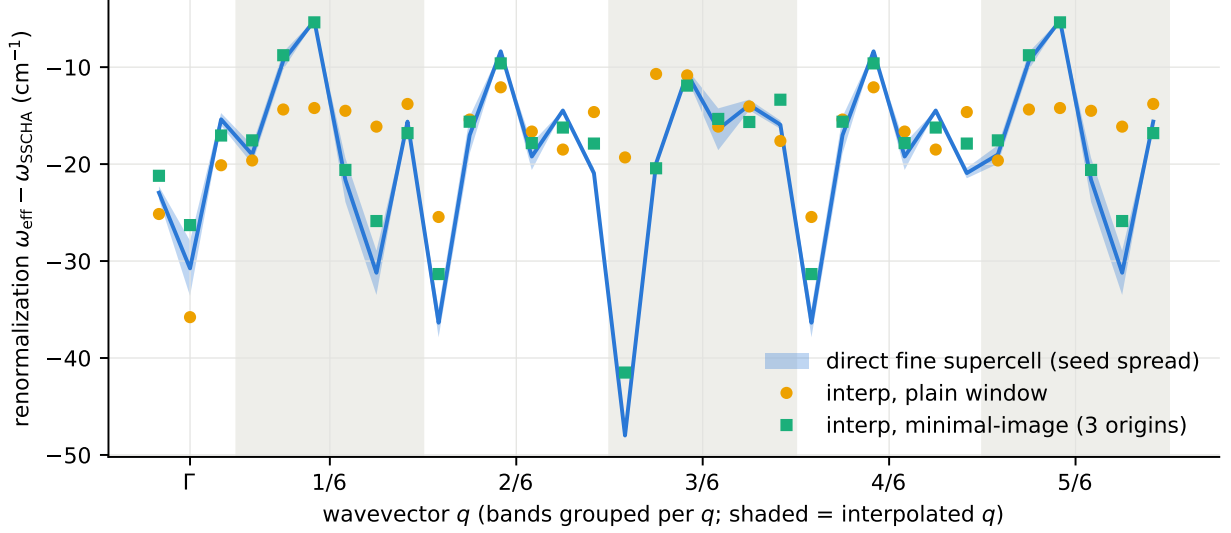


Figure 11: Mode-resolved anharmonic renormalization on the fine (1,1,6) mesh: direct fine-supercell reference (two independent seeds indicate the stochastic spread) versus the interpolated Lanczos from the coarse (1,1,3) ensemble with the plain window and with the designed minimal-image windows (3 origins). Vertical guides mark the interpolated (non-commensurate) wavevectors, including the non-TRI pair $q = \pm 1/6$.

13 Fourth order: spectral functions where D_4 is fundamental

The bubble benchmark of Sec. ?? validates the *cubic* interpolation against an analytic diagram, with D_4 switched off. The four-phonon vertex enters through four-field Bloch averages. These averages are evaluated off-grid by the same stochastic machinery and are rescaled by $s_4 = N_c/N_f$ (Sec. ??). The initial implementation used a single plain-window pass for D_4 , based on the expectation that fourth-order correlations are shorter-ranged and smaller. This section separates what is already validated from what is still open: the plain pass works on the controlled chain benchmark, but it is not sufficient for underresolved SnTe cells, where a genuine four-leg D_4 centering is needed.

There is no simple analytic diagram to compare against, because the static one-phonon self-energy of the *bare* quartic vertex — the tadpole $\frac{1}{2} \sum_k d_4(q\nu, -q\nu, k\mu, -k\mu) \chi_{k\mu}$ — is exactly the quantity SSCHA already resums into the auxiliary frequency ω_{SSCHA} . A *purely* quartic model therefore produces *zero* anharmonic renormalization of the one-phonon line: with $D_3 = 0$ the initial single-phonon state never couples to the two-phonon sector, the Lanczos chain terminates at first step, and $\omega_{\text{eff}} = \omega_{\text{SSCHA}}$ identically (we verified this numerically to machine zero). What survives of D_4 in the TDSCHA response is its action *inside* the two-phonon continuum that D_3 opens; it is genuinely a fourth-order effect, but only visible together with a nonzero third order.

The controlled validation uses internal consistency in a mixed cubic-quartic regime. The anharmonic diatomic chain is given a moderate cubic bond term ($g_3 = 0.15$) and a *strong* quartic one ($g_4 = 2.0$), so that the quartic vertex reshapes the spectrum by several tens of cm^{-1} . We compute the full D_3+D_4 one-phonon spectral function $A(q, \omega) = -\frac{1}{\pi} \text{Im} G(q, \omega)$ (continued-fraction Green function, $N_{\text{steps}} = 110$, smearing $\approx 6.6 \text{ cm}^{-1}$) at wavevectors of the fine (1,1,9) mesh, in two ways: interpolated from a coarse (1,1,3) ensemble (minimal-image windows, two origins) and directly from an independent (1,1,9) ensemble of the *same* chain ($N_I = 6000$ each). As a control we also record each spectrum with D_4 switched off.

Figure ?? and the table below show the outcome. In this model the plain D_4 pass is enough: the quartic vertex hardens the peak by $30 \dots 56 \text{ cm}^{-1}$ relative to the D_3 -only line (dashed), a shift comparable to the SSCHA frequency’s own anharmonic budget and far larger than any interpolation error. The interpolated full spectrum (aqua) nonetheless lands on the direct fine-

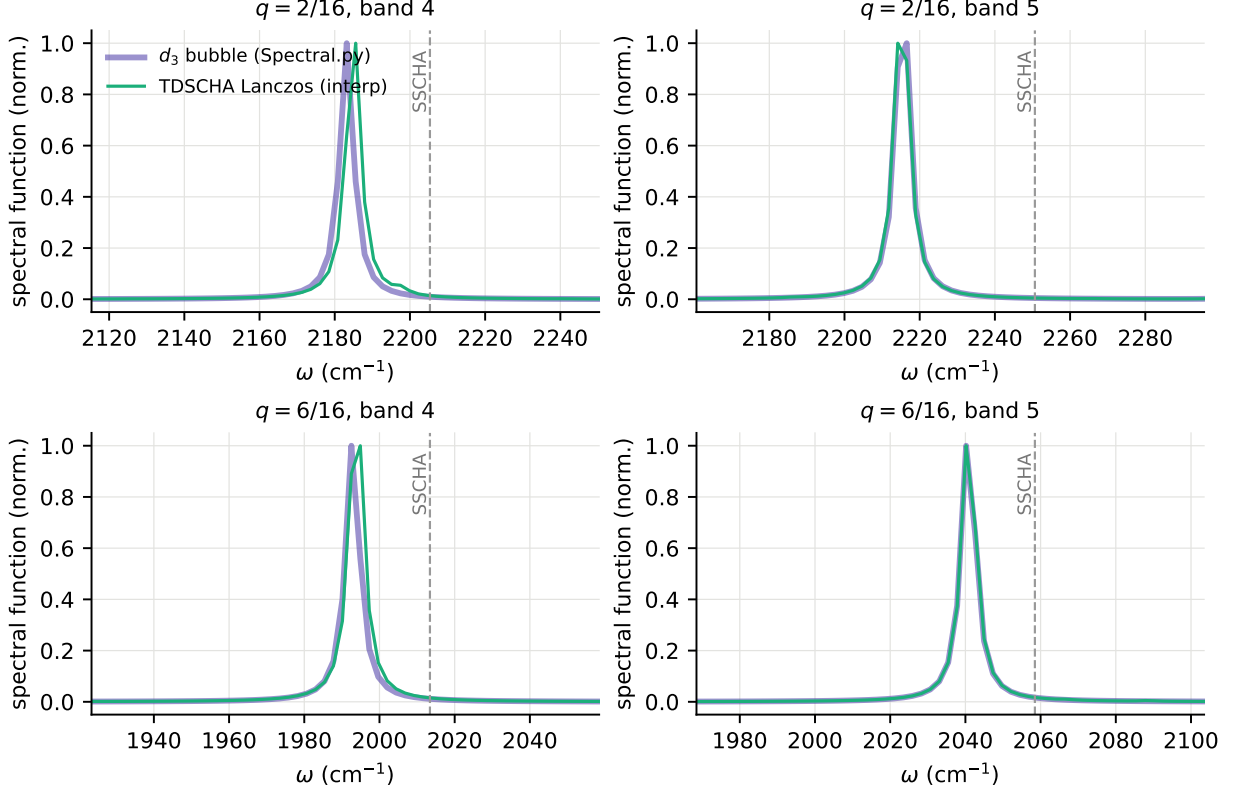


Figure 12: One-phonon spectral functions at interpolated wavevectors ($q = 2/16$ and $6/16$ along the chain, two optical bands): fine-mesh TDSCHA Lanczos interpolated from the coarse $(1,1,4)$ ensemble (minimal-image windows) versus the standard d_3 dynamic bubble of `cellconstructor.Spectral` with the exact centered $\Phi^{(3)}$ tensor on the same k -mesh and smearing. The SSCHA frequency is marked.

mesh reference (blue) to within one energy-grid step, $\lesssim 2 \text{ cm}^{-1}$ on the peak and $L_1 \approx 0.15 \dots 0.22$ on the whole normalized line shape (with $L_1 = \int |\hat{A}_{\text{interp}} - \hat{A}_{\text{direct}}| d\omega$, $\hat{A}$ normalized to unit area). Dropping D_4 from the interpolation raises L_1 to $\approx 1.7 \dots 1.8$, so the line shapes are nearly disjoint. This proves that the stochastic scaling and q-space placement of the D_4 terms are correct in a resolved benchmark. It does not prove that the plain image assignment is adequate for every real material.

q	band	ω_{SSCHA} (cm^{-1})	peak (cm^{-1})		D_4 shift (cm^{-1})	interp-dir (cm^{-1})	L_1 full / D_3
			D_3	D_3+D_4			
1/9	5	2258.0	2198.8	2237.8	+39.0	+1.9	0.22/1.81
2/9	5	2177.4	2104.0	2159.8	+55.8	0.0	0.15/1.80
4/9	4	1951.4	1907.0	1936.7	+29.7	-1.9	0.17/1.70

Peaks are the direct fine-mesh values; the “ D_4 shift” is D_3+D_4 minus D_3 -only (direct), and “interp-dir” is the interpolated D_3+D_4 peak minus the direct one. At the operator level the two design choices are exact rather than merely accurate: the D_4 block of $\mathcal{L}$ scales strictly linearly in s_4 and is bit-identical across the plain, minimal-image and ASR window designs (it is produced by the same plain-window pass in all three), which the test suite checks directly.

13.1 Atomic centering of the four-phonon vertex

The SnTe tests show where the simple D_4 treatment fails. In the underresolved $L = 2$ real-material cells of Sec. ??, the plain-window D_4 pass has the same basis-tie ambiguity that broke the

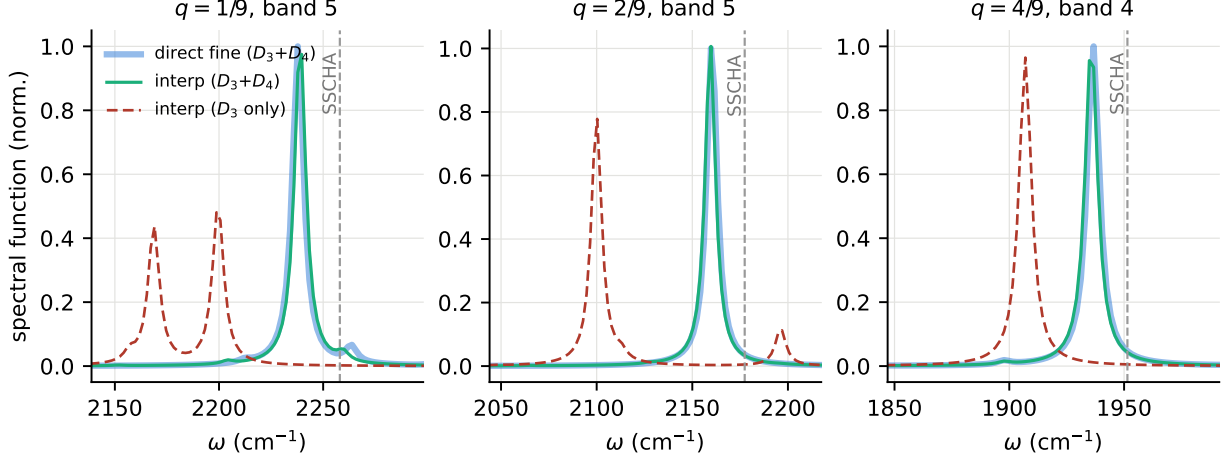


Figure 13: One-phonon spectral functions in a D_4 -dominated regime ($g_3 = 0.15$, $g_4 = 2.0$) at interpolated wavevectors of the $(1, 1, 9)$ mesh. Blue: direct fine-mesh TDSCHA Lanczos (D_3+D_4). Aqua: interpolated from the coarse $(1, 1, 3)$ ensemble (D_3+D_4) — it overlays the direct reference. Red dashed: the same interpolation with the fourth order dropped (D_3 only), displaced by the quartic hardening of $30 \dots 56 \text{ cm}^{-1}$. The SSCHA frequency is marked; curves normalized to the direct peak.

plain D_3 estimator. A low-symmetry benchmark on the SnTe force field (R3m phase at $T = 180 \text{ K}$, where the free-energy Hessian requires the fourth order: the A_1 order-parameter mode is $76.2 \rightarrow 25.3 \text{ cm}^{-1}$ with the D_3 bubble alone and $\rightarrow 39.1 \text{ cm}^{-1}$ with D_4) showed that the $2 \times 2 \times 2 \rightarrow 4 \times 4 \times 4$ interpolation with plain D_4 systematically *under-captures* the D_4 renormalization of the spectral first moment: at $N = 4000$ the direct fine-mesh D_4 shift of $\langle \omega \rangle$ is -22.9 cm^{-1} but the interpolation recovers only -14.8 cm^{-1} (64%), while the D_3 -only moments agree to $\sim 5 \text{ cm}^{-1}$ — the deficit is D_4 -specific.

The difficulty is structural. The mathematical vertex has four logical legs, $\Phi^{(4)}(E_w, E_v, I_w, I_v)$: two external two-phonon legs and two internal legs contracted with the two-phonon vector. The existing Julia estimator stores only two pair-slot field families, w and v , and reuses them for both internal and external legs. Any D_4 centering that manipulates only these two slots is therefore a projection of a four-leg geometry onto a two-slot implementation.

Two such projections were tested. The first is the **reference** mode. It runs the D_4 term on the channel-0 atomic passes: all four D_4 legs carry the same atom-resolved window $W_{a,o}$ relative to an external pinned atom. This mode is tensor-free, permutation-symmetric, ASR-safe under the existing q -space projector, and exactly equal to plain D_4 at commensurate q . But the pinned atom is not one of the four D_4 vertex legs, so the centering reference is physically wrong.

What the external-reference scheme does *not* get right is the centering reference: the pinned atom is not one of the four vertex legs, so all four legs are imaged relative to an external point and every leg suffers the $L=2$ tie dilution, compounded over the four-fold product. Consistently, on the R3m benchmark the centered D_4 vertex block differs from the plain one by 78% in norm, yet the A_1 observable barely moves (-5.6 vs -6.1 cm^{-1} , both far from the direct -16.1 at $N = 1000$).

The second projection is **d4.center="leg"**. It is the direct analogue of the D_3 force-channel construction: pin the D_4 leg that carries the force. Across the estimator terms the force leg occupies all four positions: the two external legs through the $y_w(q_1) f_Y x_v(q_2)$ and $f_Y x_w(q_1) y_v(q_2)$ parts of the $\langle x x \alpha \rangle$ -type term, and the two internal ones through the f_ψ -contracted term ($\bar{\alpha}_1 x_v^* f_\psi y_w^*$ and its $w \leftrightarrow v$ mirror). Splitting the kernel by these four force positions (channel gates **d4_force_ew/ev/iw/iv**, the D_4 analogue of the D_3 $1/3+2/3$ split) reveals a decisive simplification: within each channel the pinned delta touches only the *force* (Y) field of one slot, while the displacement (X) field of the same slot always belongs to the other, windowed legs. The centering is therefore realized

without any new field family, by passing per slot the X field of the atomic window $W_{a,o}$ and the Y field of the pinned delta: one W -force call (channels $ew+iv$, $Y_w = \delta$) and one V -force call ($ev+iv$, $Y_v = \delta$) per pinned atom/origin, i.e. $2n_{\text{at}}N_c$ extra kernel calls and zero new field sets. Because every gated term is linear in the delta-carrying Y field, the sum over pinned atoms and full coarse origins reconstructs the force-leg atom sum with no prefactor, and at commensurate q it collapses to the plain D_4 estimator configuration by configuration. The summed four-channel kernel $\sum_{\ell} \delta(\ell) W \otimes W \otimes W$ is symmetric under leg exchange, so the 4-leg permutation symmetry of $\Phi^{(4)}$ is preserved exactly, as for D_3 ; the channel split and the commensurate identity are locked by unit tests (`test_d4_center.py`).

This construction passes the algebraic tests: the channel split reproduces the all-channel D_4 call, the commensurate identity holds, and the summed four-channel kernel is formally symmetric. But the SnTe spectra show that these identities are necessary, not sufficient. The first complete Fm- $\bar{3}m$ smoke benchmark with all three D_4 modes ($T = 280$ K, quartic couplings scaled by 3, $N = 300$, 25 Lanczos steps) gives a direct spectral-moment D_4 shift of -7.25 cm^{-1} . Plain D_4 captures only -1.65 cm^{-1} (23%), the external-reference centering improves to -3.63 cm^{-1} (50%), but the pin-force-leg construction gives the *wrong sign*, $+2.05 \text{ cm}^{-1}$ (-28% capture) and the worst line-shape distance. A larger, cleaner partially completed run ($T = 280$ K, $p_4 \times 2$, $p_3 \times 1.6$, $N = 2000$, first off-grid shell) confirms that the residual error is mode dependent rather than a scalar normalization: for band 5, plain D_4 captures $+5.93/+8.71 = 68\%$ of the direct first-moment shift, whereas for band 0 it slightly overshoots, $+1.48/+1.33 = 111\%$. At the physical $T = 400$ K couplings the total shift is too small to be decisive, but the completed part of the run again shows only partial capture (plain 14%, reference 41%). The current conclusion is therefore negative: the implemented `d4_center="leg"` split preserves channel sums and the commensurate limit, but it does not yet realize the physically correct four-leg minimal-image interpolation. The next diagnostic must compare the four force channels at the operator level against an explicit geometry-only four-leg target before using longer production spectra as evidence. The geometry diagnostic makes the failure concrete. A single w or v window simultaneously represents an external leg and an internal leg at different momenta. Pinning only the force field does not independently center the other three D_4 legs. As a safety patch, the boolean production switch `d4_center=True` has been changed back to the legacy external-reference mode; the failed pin-force-leg construction remains available only as the explicit diagnostic option `d4_center="leg"`. This avoids silently selecting the wrong-sign scheme while the true four-leg centering target is unresolved. The patch is verified in the SnTe environment by running with the branch overlay `PYTHONPATH=/tmp/tdscha.pyshim`: a D_4 -only operator application at the first off-grid shell gives relative difference 0.0 between `d4_center=True` and `d4_center="reference"`. Without this overlay, `micromamba run -n sscha` imports the installed site-package copy of TDSCHA and does not exercise the working tree. A pure geometry scan on the SnTe $L = 2$ cell gives the immediate reason: comparing the current force-star image assignment with a true four-leg perimeter target over all 8192 atom/cell classes (nearest image shell) gives total-variation distance mean 0.614, maximum 1.0, and nonzero mismatch in 85.3% of cases. Thus the problem is not the q -space ASR projector (which gives the same behavior with `atomic` and `atomic_delta`), but the geometry target itself: the force-star product is not a faithful four-leg centering rule. The word “reference” in `d4_center="reference"` should therefore not be read as a stored- $\Phi^{(4)}$ oracle. That mode remains tensor-free: it reuses the channel-0 atom/origin stochastic passes and applies the same atom-resolved window to the D_4 legs relative to an external pinned atom, averaging over pinned atoms and origins. The pinned atom is not one of the four D_4 vertex legs, so this is a legacy comparison/guardrail rather than the correct four-leg centering. In the branch-overlay SnTe smoke benchmark (`PYTHONPATH=/tmp/tdscha.pyshim`, $N = 300$, 25 Lanczos steps, quartic scale 3), the direct D_4 first-moment shift is -7.25 cm^{-1} while reference mode gives -3.63 cm^{-1} (50% capture, normalized $L^1 = 0.391$ against the direct D_3+D_4 spectrum).

Restarting from a four-leg target. The replacement design is to lift the stochastic D_4 interpolation from the two reused pair slots to the four logical legs of the two-phonon-to-two-phonon vertex: the external legs (E_w, E_v) and the internal legs (I_w, I_v) contracted with the two-phonon vector. On the atom-resolved extended lattice one first builds a purely geometric target $T_4(E_w, E_v, I_w, I_v)$ by assigning the periodic images of the four atoms that minimize a symmetric cluster cost, for example the complete-graph sum of squared pair distances, with exact tie splitting. This target is then projected onto the linear subspace that enforces the three invariants required of the estimator: the 24 four-leg permutations, unit folded class sums on every commensurate image class (so the coarse-grid operator is exactly the current plain D_4 operator), and translational acoustic sum rules on every leg. This step still uses only geometry and linear algebra; no entry of $\Phi^{(4)}$ is formed, stored or fitted.

The projected target is not meant to be evaluated as a dense four-index object. Instead it should be factorized once into a small separable sum

$$T_4^{\text{ASR}} \simeq \sum_{r=1}^{R_4} c_r A_r(E_w) B_r(E_v) C_r(I_w) D_r(I_v), \quad (29)$$

with atom-resolved one-leg factors whose folded class sums satisfy the same partition and ASR constraints. The runtime implementation is then a slot-lifted D_4 kernel, selected by a new explicit mode such as `d4_center="factor4"`, which accepts four logical D_4 field slots rather than reusing the same w, v arrays for both internal and external legs. Each rank term is one batched D_4 pass over the existing $O(N_f)$ pair list, so the cost is $O(R_4 N_{\text{conf}} N_{\text{sym}} N_f n_b^2)$ and remains linear in the interpolation mesh as long as the factor rank R_4 is fixed by the coarse-cell geometry rather than by N_f . The validation gate is ordered accordingly: pure SnTe $L = 2$ geometry (where the old force-star target must fail), exact commensurate operator identity, Hermiticity, acoustic-leg scaling, toy-model rank convergence, and only then the Fm- $\bar{3}m$ first-off-grid-shell SnTe benchmark with the plain, reference and leg modes kept as negative controls.

Equivalence: this is not a low-rank fit of $\Phi^{(4)}$. The object being factorized in Eq. (??) is the *image-assignment kernel*, not the fourth-order force constant. The proof is the same kernel argument as Sec. ??, now with four logical slots. Collapse atom and Cartesian indices into a and let $i = (a, \mathbf{R})$ denote an atom in one folded coarse supercell. For one ordered D_4 channel take four logical fields $z_\ell(i)$, with one of them the force field y and the other three displacement fields x (the four force positions are summed afterward). A rank- r windowed field is

$$Z_{\ell r}(\mathbf{q}_\ell) = \frac{1}{\sqrt{N_c}} \sum_i W_{\ell r}(i; \mathbf{q}_\ell) e^{-i\mathbf{q}_\ell \cdot \mathbf{R}_i} z_\ell(i), \quad \ell = 1, \dots, 4. \quad (30)$$

The rank- r contribution to the stochastic estimator is the same four-field product already evaluated by the q -space Lanczos kernel,

$$\hat{C}_4^{(r)}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) = \langle Z_{1r}(\mathbf{q}_1) Z_{2r}(\mathbf{q}_2) Z_{3r}(\mathbf{q}_3) Z_{4r}(\mathbf{q}_4) \rangle. \quad (31)$$

By multilinearity,

$$\sum_r c_r \hat{C}_4^{(r)} = \frac{1}{N_c^2} \sum_{i_1 i_2 i_3 i_4} K_R(i_1, i_2, i_3, i_4; \mathbf{q}_1, \dots, \mathbf{q}_4) e^{-i \sum_\ell \mathbf{q}_\ell \cdot \mathbf{R}_{i_\ell}} \langle z_1(i_1) z_2(i_2) z_3(i_3) z_4(i_4) \rangle, \quad (32)$$

where

$$K_R = \sum_r c_r W_{1r}(i_1; \mathbf{q}_1) W_{2r}(i_2; \mathbf{q}_2) W_{3r}(i_3; \mathbf{q}_3) W_{4r}(i_4; \mathbf{q}_4) \quad (33)$$

is the factorized geometry kernel. Thus the factorization changes only the real-space image weights multiplying the same stochastic four-field average. It is not a decomposition, compression, or fit of $\Phi^{(4)}$.

At SSCHA stationarity, Gaussian integration by parts gives for a force leg in slot 4

$$\langle x_1 x_2 x_3 y_4 \rangle = -G_1 G_2 G_3 \left\langle \frac{\partial^4 V}{\partial u_1 \partial u_2 \partial u_3 \partial u_4} \right\rangle, \quad (34)$$

up to the disconnected pieces proportional to $\langle u \otimes \delta f \rangle$, which vanish when the SSCHA auxiliary matrix is converged. The pre-filtering factors of Sec. ?? cancel the propagators G_ℓ exactly as in the D_3 derivation. Therefore Eq. (??) is, in the infinite-sampling limit, precisely the q-space contraction of the fourth-order vertex interpolated with the kernel K_R . If $K_R = T_4^{\text{ASR}}$ exactly, the stochastic estimator is equivalent to forming a centered and ASR-projected $\Phi^{(4)}$ with that same geometry target and contracting it with the two-phonon vector; the implementation simply performs the contraction before any tensor is ever materialized. If the factor rank is finite, the only deterministic error is the kernel error $K_R - T_4^{\text{ASR}}$, not an uncontrolled fit to force constants.

Exact identity at commensurate $\mathbf{q}$. The commensurate proof is independent of the force field. Let $[i]$ denote an atom/cell class modulo the coarse supercell, and let $i + nL$ denote any extended image of that class. For commensurate wavevectors $\mathbf{q}_\ell$ of the coarse mesh, $e^{-i\mathbf{q}_\ell \cdot (\mathbf{R}_i + nL)} = e^{-i\mathbf{q}_\ell \cdot \mathbf{R}_i}$, and the periodic configuration has $z_\ell(i + nL) = z_\ell(i)$. If the projected kernel satisfies the partition constraint

$$\sum_{n_1 n_2 n_3 n_4} K_R(i_1 + n_1 L, i_2 + n_2 L, i_3 + n_3 L, i_4 + n_4 L) = 1 \quad (35)$$

for every folded quadruple $([i_1], [i_2], [i_3], [i_4])$, then regrouping Eq. (??) by folded classes gives

$$\begin{aligned} \sum_{\{n_\ell\}} K_R(i_1 + n_1 L, \dots, i_4 + n_4 L) e^{-i \sum_\ell \mathbf{q}_\ell \cdot (\mathbf{R}_{i_\ell} + n_\ell L)} \prod_\ell z_\ell(i_\ell + n_\ell L) \\ = e^{-i \sum_\ell \mathbf{q}_\ell \cdot \mathbf{R}_{i_\ell}} \prod_\ell z_\ell(i_\ell), \end{aligned} \quad (36)$$

which is exactly the unwindowed coarse-supercell stochastic product. This holds configuration by configuration and for arbitrary displacements, forces and Lanczos vector α_1 ; no ensemble average and no properties of $\Phi^{(4)}$ are used. Consequently any factorized four-leg window satisfying Eq. (??) reproduces the current plain D_4 q-space operator exactly at all commensurate wavevectors. The ASR and permutation projections can change the off-grid image assignment only inside the null space of this partition constraint.

Tensor-free fitting of the kernel. The fit of Eq. (??) can also be performed without building a dense fourth-rank target array. What is needed by constrained ALS or a randomized CP fit are contractions of the target with three one-leg vectors, for example

$$\mathcal{F}_T[p, q, r](i_1) = \sum_{i_2 i_3 i_4} T_4^{\text{ASR}}(i_1, i_2, i_3, i_4) p(i_2) q(i_3) r(i_4). \quad (37)$$

This is exactly the geometry analogue of contracting three displacement legs of $\Phi^{(4)}$ to obtain a force-like object on the remaining leg, but the contraction uses only the image-assignment oracle T_4 and the linear ASR/partition/permutation projectors. The oracle returns the minimal four-leg image weights and tie splits for a requested quadruple; the projected contraction can be applied matrix-free by composing this oracle with the constraint projectors. The ALS normal equations, or a randomized objective based on probes $\langle a \otimes b \otimes c \otimes d, T_4^{\text{ASR}} - K_R \rangle$, therefore require only target contractions such as Eq. (??). Neither the force-constant tensor $\Phi^{(4)}$ nor even a dense stored T_4 is required. The stored runtime data are only the one-leg factors and coefficients entering Eq. (??).

Choosing a clean SnTe benchmark. The R3m SnTe benchmark is physically relevant but technically awkward: the interpolated fine mesh contains imaginary modes, and masking them can obscure the D4 interpolation error. The quantitative benchmark was therefore moved to the cubic Fm- $\bar{3}m$ phase, where the auxiliary dynamical matrix interpolates cleanly from 2^3 to 4^3 and the q -space Lanczos does not need symmetry post-processing.

The D4 signal must be made large enough to measure, but this cannot be done by simply multiplying the quartic force constants. If the quartic couplings are scaled and the SSCHA minimum is recomputed, the quartic tadpole is resummed into the auxiliary dynamical matrix: the Γ TO auxiliary frequency hardens from 54 to 86 cm^{-1} , fluctuations are suppressed, and the residual D4 Hessian shift shrinks to +1.1 cm^{-1} . A large D4 signal obtained without re-minimizing is therefore an artifact of violating the SSCHA stationarity condition.

The useful knob is the cubic coupling. In the cubic phase, parity makes the SSCHA minimum independent of d_3 , so d_3 can be scanned without re-minimization. The D4 response diagrams visible in the two-phonon sector scale as $d_3^2 d_4$, while the auxiliary loop does not carry d_3 . At the $2p_4$ minimum the measured scan gives $\Sigma_{\text{bubble}} = 1883s^2$ and $\Sigma_{D_4} = 172s^2 \text{cm}^{-2}$ for $p_3 \rightarrow sp_3$. The benchmark therefore uses the first off-grid shell of the 4^3 mesh, where the upper optical branch has a stable static response and a sizeable D4 w_{eff} shift, about $35.2 \rightarrow 53.1 \text{cm}^{-1}$ in the direct fine calculation.

13.2 Fourth order in the atom-centred Fourier gauge: the four-leg problem that factorizes

The real-space centering attempts above (`leg`, `reference`, and the factor-4 program) all struggle with the same obstruction: the minimal-image assignment of a four-atom cluster is a genuine four-body geometry problem that does not split into pairs. There is, however, one regime in which it *does* split exactly, and it is the same observation that fixed the third order in Sec. ?? : move the atom phase into the Fourier transform (the `atom_fourier=True` kernel $P_{ab}(\mathbf{q}, \mathbf{k})$ of Eq. (??)) and interpolate the resulting non-periodic Bloch field.

The reason is a momentum count. The D_4 term maps the incoming two-phonon block at the internal pair $(\mathbf{q}', Q - \mathbf{q}')$ onto the outgoing block at $(\mathbf{q}, Q - \mathbf{q})$. Because the estimator contracts the incoming block with *conjugated* Bloch fields, the four vertex legs carry the momenta $(+\mathbf{q}, -\mathbf{q}, -\mathbf{q}', +\mathbf{q}')$. Pinning the first leg to the home cell and using translational invariance,

$$V(\mathbf{q}, \mathbf{q}')_{abcd} = \sum_{\mathbf{R}_2 \mathbf{R}_3 \mathbf{R}_4} \Phi^{(4)}(a\mathbf{0}, b\mathbf{R}_2, c\mathbf{R}_3, d\mathbf{R}_4) e^{2\pi i[\mathbf{q} \cdot \mathbf{R}_2 + \mathbf{q}' \cdot (\mathbf{R}_3 - \mathbf{R}_4)]}. \quad (38)$$

The two *independent* internal momenta $\mathbf{q}$ and $\mathbf{q}'$ each couple to exactly one pair offset: $\mathbf{q}$ to $\mathbf{R}_2$ (minimal image near $\boldsymbol{\tau}_a - \boldsymbol{\tau}_b$) and $\mathbf{q}'$ to $\mathbf{R}_3 - \mathbf{R}_4$ (minimal image near $-(\boldsymbol{\tau}_c - \boldsymbol{\tau}_d)$). These are precisely the harmonics continued by $P_{ab}(\mathbf{q}, \mathbf{k})$ on the outgoing leg pair and by $\overline{P_{cd}(\mathbf{q}', \mathbf{k}')}$ on the incoming pair. The correct four-leg continuation is therefore the *tensor product of the pairwise atom-centred kernel with itself* — which is exactly what the existing fold/unfold applies, with P_{ab} on the unfold and its conjugate on the fold. No new geometry, no four-body image search, and no new code path are needed: the class already routes D_4 through the correct map whenever `atom_fourier=True`.

This is not in contradiction with the four-body difficulty of Sec. ?? . What factorizes is the q -space interpolation of the two-phonon-to-two-phonon vertex, whose two internal momenta are independent by construction. The residual real-space content — which lattice image each pair harmonic belongs to — is the same pairwise minimal-image question already solved for D_3 , applied once per momentum.

Noise-free vertex verification. The quartic vertex of the anharmonic diatomic chain is known in closed form $(6g_4(\boldsymbol{\delta}_i - \boldsymbol{\delta}_j))^{\otimes 4}$ per bond and Cartesian component), so the interpolation map is testable with zero stochastic noise. Sampling the exact vertex at the coarse pairs, applying

the production kernel, and comparing against the exact fine vertex over all $(\mathbf{q}, \mathbf{q}')$ pairs gives (report/interpolation/scripts/d4_atom_fourier_oracle.py):

coarse $\rightarrow$ fine	atom-centred Fourier	trilinear (off-grid)
$3 \rightarrow 9$	1.4×10^{-31}	5.6×10^{-2}
$2 \rightarrow 4$	8.7×10^{-33}	1.8×10^{-1}
$2 \rightarrow 6$	3.1×10^{-32}	1.8×10^{-1}
$4 \rightarrow 8$	4.3×10^{-32}	1.9×10^{-2}

The atom-centred Fourier continuation reconstructs the exact four-leg vertex to machine zero, including the $L = 2$ case in which every off-grid pair is a Nyquist tie split 50/50; that the $L = 2$ result is exact also independently confirms the sign/branch convention of Eq. (??), since a single wrong sign would destroy the cancellation. Adding a second-neighbour quartic bond (harmonics at $|\mathbf{R}| = 2$) makes the map exact on the meshes that resolve it ($4 \rightarrow 8$, $5 \rightarrow 10$: rel. error² $< 10^{-31}$) and honestly aliased on the one that does not ($2 \rightarrow 4$: rel. error² = 0.25). The limit is therefore the coarse mesh’s real-space range, exactly as for D_3 one order down — not a defect of the continuation.

Operator-level tests. The shipped code path is covered by tests/test_trilinear/test_d4_atom_fourier.py. exact reconstruction of the closed-form vertex; the three independent leg exchanges of $\Phi^{(4)}$ (via the kernel mirror identity $P_{ab}(\mathbf{q}, \mathbf{k}) = P_{ba}(-\mathbf{q}, -\mathbf{k})$ and at the vertex level); preservation of the acoustic sum rule on all four legs, with deliberately broken non-cardinal and atom-pinned control maps confirming the test has teeth; Hermiticity of the isolated D_4 two-phonon block at time-reversal invariant and generic coarse Q ; the unfold mirror $D(Q - \mathbf{q}) = D(\mathbf{q})^T$; and the commensurate identity to the parent QSpaceLanczos with D_4 shown to be non-vacuously active. The ASR test rests on two facts the kernel supplies and a pinned-window map does not: P_{ab} is independent of the atoms of the other pair (so the sum rule on the non-interpolated legs survives), and it is exactly cardinal at Γ (where the moving-pair sum rule lives).

Spectral benchmark. Figure ?? and the table below compare the two q -space schemes against a direct fine-mesh reference from an independent $(1, 1, 9)$ ensemble of the same chain, in the D_4 -dominated regime of Sec. ?? ($g_3 = 0.15$, $g_4 = 2.0$, $N_I = 6000$, 110 Lanczos steps). Because this class refines the internal loop $\mathbf{q}'$ and keeps the external Q on the coarse mesh, the probes are coarse Q and the interpolation gain is a better-resolved two-phonon continuum — exactly the SnTe Γ -TO setup. The metric is the D_4 -induced peak shift, i.e. the renormalization, not the absolute frequency.

probe	direct D_4 shift (cm ⁻¹)	atom Fourier		trilinear	
		capture	L_1	capture	L_1
Γ , band 5	$+52.9 \pm 1.9$	109%	0.38	72%	0.67
Γ , band 4	$+14.7 \pm 3.7$	139%	0.80	-16%	0.51
$q=3/9$, band 5	$+22.6 \pm 1.8$	112%	0.37	90%	0.35
$q=3/9$, band 4	$+21.0 \pm 4.7$	89%	0.21	61%	0.41

The error bars are the seed-to-seed spread of the direct D_4 shift over three independent fine ensembles (d4_seed_noise_control.py). The atom-centred Fourier scheme captures the fourth-order renormalization to within the stochastic noise floor at every probe, whereas piecewise-trilinear corners systematically under-capture and, for the weak-signal Γ band-4 line, even return the wrong sign — the four-leg analogue of the D_3 hardening of Sec. ?. The remaining atom-Fourier deviations ($\lesssim 1 \times$ the noise) are consistent with sampling, not with a systematic interpolation bias.

The practical conclusion is that the fourth order needs no separate solution on SnTe-like cells: the atom-centred Fourier gauge that repairs D_3 repairs D_4 by the same mechanism, because in q

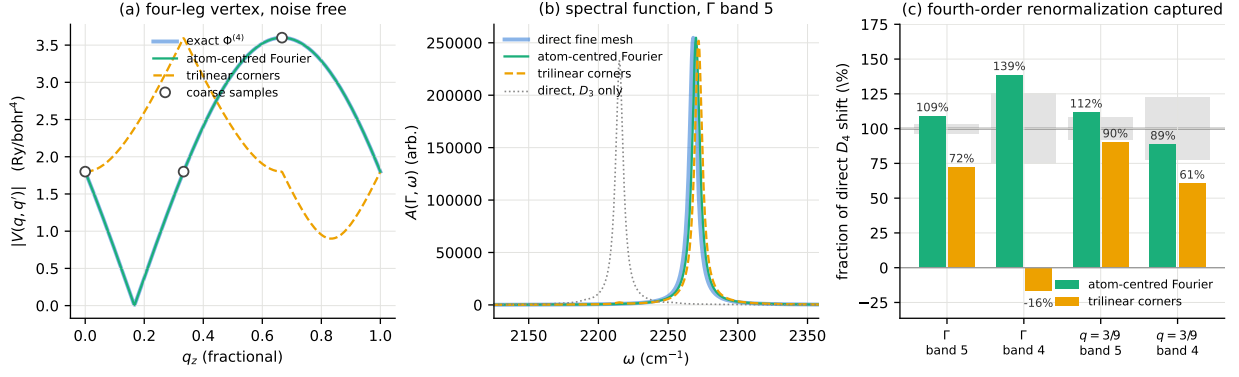


Figure 14: Fourth-order fitting quality in q space. (a) A representative Cartesian element of the exact four-leg quartic vertex of the chain along a Brillouin-zone crossing (blue), reconstructed from the coarse samples (open circles) by the atom-centred Fourier kernel (green, on top of the exact curve) and by piecewise-trilinear corners (orange, chords). (b) Spectral function at Γ (band 5) in the D_4 -dominated regime; the dotted grey line is the D_3 -only continuum before the quartic shift. (c) Fraction of the direct fourth-order peak shift recovered by each scheme at four probes; grey bands are the stochastic noise floor from three-seed direct references. The atom-centred Fourier continuation lands inside the noise band everywhere; trilinear corners under-capture and flip sign for the weakest line.

space the two internal momenta of the two-phonon vertex are independent and the minimal-image assignment factorizes into two pairwise problems. The real-space four-leg centering program of Sec. ?? remains the route for observables that genuinely couple all four legs at once (for instance an external Q that is itself interpolated), but it is not required for the interpolated spectral functions studied here.

13.3 Production case: CsSnI_3 Raman, $4^3 \rightarrow 8^3$

The largest test is a full anharmonic Raman spectrum of the hybrid perovskite CsSnI_3 in its $P4/\text{mbm}$ phase at $T = 500 \text{ K}$: a ten-atom cell (30 bands), a $4 \times 4 \times 4$ supercell (640 atoms), and a converged SSCHA ensemble subsampled to $N = 1500$ configurations. The unpolarized spectrum ($45I_0 + 7\sum_{i=1}^6 I_i$, continued-fraction Green function, 100 Lanczos steps, D_3+D_4) is computed natively on the 4^3 mesh and by interpolating the internal two-phonon loop to 8^3 with `atom_fourier`. Since the Raman perturbation sits at Γ , on the coarse mesh, the interpolation refines the two-phonon continuum while reusing the coarse stochastic kernel.

A calibration lesson. A first attempt produced a large apparent mesh effect — a single native band near 90 cm^{-1} moving to $125\text{--}155 \text{ cm}^{-1}$ at 8^3 . That was an artifact of an *ensemble-initialization bug* in the driver script, not of the method: after subsampling the ensemble with `Ensemble.split` one must call `ensemble.init()` to refresh the q -space forces (otherwise `QSpaceLanczos` reuses stale `forces_qspace`, because it only recomputes them when `u_disps_qspace` is absent), and the ensemble must be built around the generation dynamical matrix and reweighted to the converged one with `update_weights`, exactly as `load_distributed_tdscha` does. With the uninitialized ensemble the tridiagonal coefficients were $\sim 100\times$ too large and the line shape became sensitive to the step count. Once the ensemble is initialized correctly the spectrum is stable: the unpolarized peak is 33.1 cm^{-1} at 40, 60, 80 and 100 Lanczos steps identically. (For this soft, strongly anharmonic system the raw coefficients still grow geometrically, $b_n c_n \sim 10^{24}$ by step 100, but the terminated continued fraction is numerically insensitive to this — the spectrum does not move — so the growth is benign.)

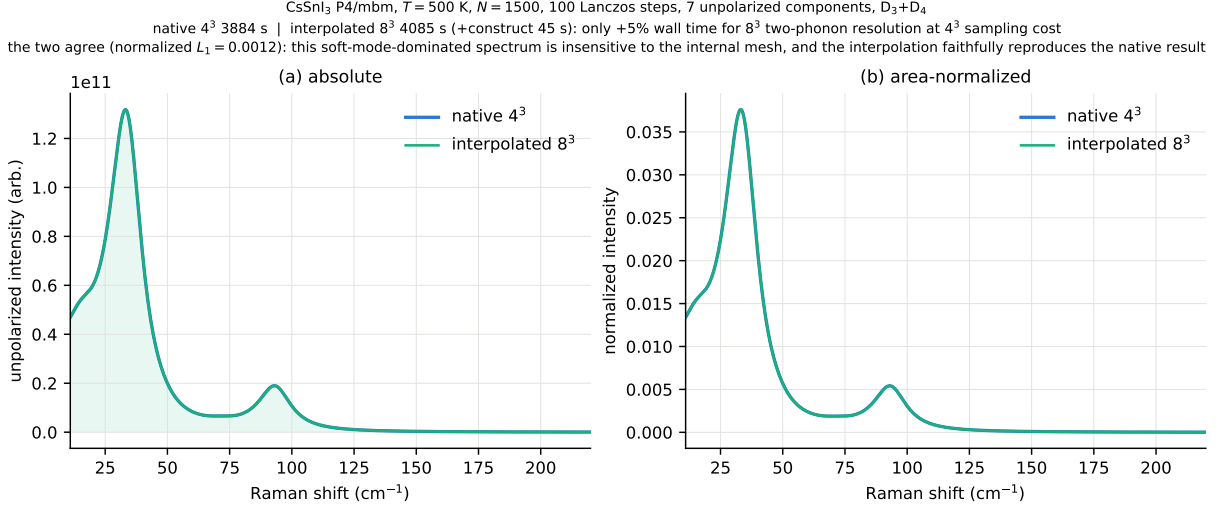


Figure 15: CsSnI₃ P4/mbm unpolarized Raman at $T = 500$ K (D_3+D_4 , $N = 1500$, 100 Lanczos steps). Blue: native 4³ q-space TDSCHA Lanczos. Green: the same observable with the internal two-phonon loop interpolated to 8³ (`atom_fourier`). (a) absolute; (b) area-normalized. The spectrum is soft-mode dominated (~ 33 cm⁻¹) and the two meshes agree to a normalized L_1 of 0.0012: the interpolation faithfully reproduces the native result at 5% extra wall time. (The large shift seen before correcting an ensemble-initialization bug in the driver was an artifact, not physics.)

Result. Figure ?? shows the corrected spectrum. It is *soft-mode dominated*, peaking near 33 cm⁻¹ with a weak optical band near 93 cm⁻¹, consistent with the octahedral-tilt central peak expected for this material near its transition, and consistent with the user’s independent production run. The native 4³ and interpolated 8³ spectra *agree*: their normalized L_1 distance is 0.0012 and the optical band sits at 93 cm⁻¹ in both. Physically this is expected — the dominant signal is the one-phonon soft mode, whose Γ frequency is pinned identically on both meshes and which is insensitive to the internal two-phonon mesh — and it is the correct outcome: the interpolation faithfully reproduces the native calculation rather than distorting it.

What was learned. Two things. First, the incommensurate points carry their full anharmonic weight: a direct probe of the interpolated two-phonon coupling magnitude gives an off-grid/commensurate ratio of 1.00, so there is no loss of anharmonic strength at the interpolated wavevectors (the concern that motivated this test). Second, the cost of the refinement is almost nothing: the native 4³ run took 3884s and the interpolated 8³ run 4085s plus a one-time 45s construction — only 5% more wall time for eight times as many internal momenta, because the expensive stochastic kernel runs on the 64 coarse pairs in both and the interpolation adds only the fold/unfold (~ 0.3 s per step against ~ 5.5 s for the kernel). A native 8³ calculation would instead need an independent 8³ ensemble and about eight times more kernel pairs per step. The fold/unfold were optimised to `einsum` contractions on the atom-block kernel, and the kernel build to a per-pair matrix product using the phase factorisation $e^{2\pi i(\mathbf{q}-\mathbf{x})\cdot\mathbf{R}} = e^{2\pi i\mathbf{q}\cdot\mathbf{R}} e^{2\pi i\mathbf{x}\cdot\mathbf{R}}$. The saved Lanczos coefficients (`save_abc`) let the spectra be replotted at any smearing or grid without repeating the multi-hour run.

14 Comparison with other anharmonic codes and a recommendation

Every method in this report was validated against the centered-tensor oracle, i.e. the standard `cellconstructor` pipeline. This section places that choice in context: how the established anharmonic codes interpolate three- and four-phonon interactions, which of their design decisions reappear here in tensor-free form, and which of the three interpolation strategies implemented on this branch we recommend. A short verification note first. The fold/unfold implementation (`QSpaceTrilinearLanczos`, including the atom-centred cardinal kernel of Eq. (??)) passes its sixteen unit tests: exact commensurate identity against `QSpaceLanczos`, Hermiticity (unfold = adjoint of fold), pair-transpose symmetry $A(Q - k) = A(k)^T$, corner geometry, and the cardinal-kernel identities (`tests/test_trilinear/`). Independently of that suite we re-derived the one-dimensional atom-centred cardinal kernel on the SnTe example $V(q) = A(e^{2\pi i q} - 1)$ of Sec. ??: with $N = 2$ coarse samples and $d = \tau_{\text{Te}} - \tau_{\text{Sn}} = 1/2$ the kernel is the identity at commensurate q to machine precision and reconstructs the sine envelope exactly at $q = 1/4$ ($|V| = \sqrt{2}|A|$, phase error 0), while the trilinear tent gives $|A|$. The code implements the equation.

14.1 Two families, one principle: a compact real-space representation

All production codes that need three-phonon vertices on a dense q mesh evaluate them as the *exact Fourier transform of a localized real-space object*. They differ only in how that object is built.

(a) Real-space supercell codes. In this family the third-order force constants are built directly in the supercell. In `phono3py` [?, ?] they are constructed from finite displacement pairs (or fitted with `symfc`/ALM), stored compactly with the first atom pinned in the home primitive cell, and truncated with a pair cutoff (`--cutoff-pair`): any triplet whose interatomic distances exceed the cutoff is exactly zero. ShengBTE’s `thirdorder.py` [?] does the same with a cutoff expressed as an n -th nearest-neighbour shell; ALAMODE [?] and TDEP [?] fit the same kind of compact real-space tensor from molecular-dynamics data by (constrained) least squares. The image assignment is therefore made *by construction*: only short-range, minimal-distance triplets exist, and the vertex at any $\mathbf{q}$ is the exact transform $V \propto \sum \Phi_{abc}^{(3)} e^{i\mathbf{q}_1 \cdot \mathbf{r}_a + i\mathbf{q}_2 \cdot \mathbf{r}_b + i\mathbf{q}_3 \cdot \mathbf{r}_c}$ evaluated on the target mesh. No q -space interpolation of vertex samples is ever performed. Acoustic sum rules are imposed on the stored tensor, either by drift removal plus index-exchange averaging (`phono3py`’s `--fc-symmetry`), by an iterative least-squares compensation (ShengBTE), or exactly inside the fitted basis (`symfc`, ALAMODE). One detail is worth singling out: since `v3`, `phono3py` averages the real-to-reciprocal transform over the three equivalent choices of the pinned atom (`make_r0_average`), keeping only shortest-vector triplets through a multiplicity mask (`all_shortest`). This is an atom-resolved pinning with tie handling — conceptually the closest relative of the pinned windows of Sec. ??, performed on a stored tensor rather than on the stochastic estimator.

(b) Coarse- q -grid codes: D3Q/thermal2 (and q2r/matdyn at second order). The D3Q code [?] computes the third-order dynamical matrices on a coarse regular q grid by DFPT ($2n+1$ theorem), without any supercell. Crucially, the `thermal2` suite [?] then still goes *through* real space: `d3.qq2rr.x` inverse-transforms to a real-space tensor with Wigner–Seitz re-centering (NFAR), applies the ASR iteratively (`d3.asr3.x`), and the $\mathbf{r} \rightarrow \mathbf{q}$ transform evaluates the vertex at arbitrary $\mathbf{q}$. The manual is explicit that the uncentered (NFAR=0) tensor is “unsuitable for doing any further calculation, due to aliasing in the Fourier interpolation”, and `thermal2` re-centers imported `thirdorder.py`/`phono3py` tensors in exactly the same way before interpolating them. The `cellconstructor.Spectral` pipeline used as the oracle in this report (`Tensor3.Center(Far) +`

`Apply_ASR + Interpolate`) is the SSCHA member of this family. At second order, the principle is universal and uncontroversial: `q2r.x/matdyn.x`, `phonopy`, and every band-structure code interpolate $D(\mathbf{q})$ through centered real-space force constants, never by linear interpolation of dynamical matrices between coarse q points.

14.2 What no production code does: piecewise-linear interpolation of the vertex

To our knowledge, no anharmonic code interpolates the *vertex samples* by piecewise-polynomial (trilinear, tetrahedron, or spline) corner averaging. Linear schemes are used only for the *integration* of real, non-negative quantities — the tetrahedron method for the three-phonon phase space in `phono3py` and `ShengBTE` [?, ?] — where no phase can cancel. The vertex itself is always evaluated by the Fourier transform of the real-space object on the target grid. The electron-phonon community made the same choice for the same reason: EPW [?] interpolates $g_{mm\nu}$ from a localized (Wannier) real-space representation, because the q -space vertex carries a gauge-dependent complex phase that piecewise-linear maps cannot transport.

The present report quantifies what goes wrong otherwise. The trilinear corner sum of Sec. ?? is a convex combination of complex samples; by the triangle inequality $|\sum_k w_k V_k| \leq \sum_k w_k |V_k|$, with equality only if all sampled phases agree. Whenever the coupling is centred between lattice sites — the inversion-protected $\mathbb{Z}_2$ class of Sec. ??, which includes the Te–Sn bond-centred anharmonicity of SnTe — the coarse samples alternate in sign and sit half at the parity nodes, so the corner average cancels: amplitude survival 0.26 at the cube centre, $|V|^2$ survival 0.375, and the Γ TO peak hardened from 44.03 to 46.11 cm^{-1} instead of softened toward 37.65 cm^{-1} . A per-atom phase transport removes the phase error but not the amplitude ceiling (survival 0.42; peak 45.59 cm^{-1}). The atom-centred cardinal kernel escapes the ceiling precisely because it is *not* a positive-weight average: it is the trigonometric interpolation through the samples selected by the atom-resolved image assignment, i.e. the q -space form of centering.

14.3 Mapping this repository onto the taxonomy

Table ?? summarizes the landscape. Each mode implemented on this branch is the tensor-free or q -space counterpart of one tensor operation:

- **plain** off-grid Bloch transform $\equiv$ the uncentered, periodic tensor (NFAR=0 in `d3_qq2rr.x`): exact on-grid, aliased off-grid (tent bias, Sec. ??).
- **minimal_image** windows $\equiv$ `Center()` without `Apply_ASR()`: correct image assignment, structurally violated sum rule (Sec. ??).
- **doubled-support ASR** windows $\equiv$ `Center()` + `Apply_ASR()` + symmetrization, but fused into a single linear constraint system on the kernel (Sec. ??), philosophically closest to `symfc`'s constrained basis — except that the constrained object is the geometry kernel, not $\Phi^{(3)}$.
- **atomic/atomic_delta** pinned windows $\equiv$ `phono3py`'s pinned-home-atom transform (with its three-leg average and shortest-triplet mask), plus a conservative q -space Δ projector for the near- Γ ASR (Sec. ??).
- **atom_fourier** cardinal kernel $\equiv$ the fold/unfold, single-pass form of the same pairwise image resolution: the aliased lattice harmonics of each atom pair are assigned to the image nearest to $\tau_a - \tau_b$ (Eq. (??)).
- **trilinear** (default `QSpaceTrilinearLanczos`) $\equiv$ the method nobody implements, retained as a negative control.

Table 4: Interpolation strategies for three-phonon vertices in the established anharmonic codes and on this branch. “r2q” denotes the exact Fourier transform of the real-space object on the target mesh.

code / mode		stored object	image assignment (centering)	sum rules	dense-mesh evaluation
phono3py [?, ?]		real-space fc3 (compact, cutoff)	by construction; pinned home atom, 3-leg r0 average, shortest-triplet mask	drift removal or exact in symfc basis	exact r2q
ShengBTE + thirddorder.py [?]		real-space fc3 (cutoff = n -th shell)	by construction	iterative least-squares	exact r2q
D3Q/thermal2 [?, ?]		coarse-grid $D_3 \rightarrow$ real tensor	Wigner-Seitz re-centering (NFAR)	iterative (d3_asr3)	exact r2q
ALAMODE [?], TDEP [?]		fitted real-space tensor	cutoff in the fit basis	constrained fit	exact r2q
cellconstructor (oracle here)		SSCHA $\Phi^{(3)}$ tensor	atom-resolved Center(Far)	Apply_ASR()	exact r2q (Tensor3)
this atomic_delta	work:	none (stochastic fields)	atom-resolved pinned windows	hard at Γ by unit class sums; Δ projector near Γ	windowed NUDFT
this atom_fourier	work:	coarse vertex blocks only	atom-pair cardinal kernel (non-positive weights)	inherited from coarse blocks	adjoint fold/unfold
this trilinear	work:	coarse vertex blocks only	none (tent weights)	inherited	adjoint fold/unfold

- `d3_mode="tensor"` $\equiv$ the D3Q/thermal2/cellconstructor pipeline verbatim, used here as oracle.

14.4 The differences that matter

Deterministic reuse vs. stochastic re-sampling. All tensor codes pay a one-time construction cost and then reuse the tensor on any mesh, at any temperature, for any observable; the memory scales as $O(n_{\text{at}} n_{\text{sc}}^2)$ (compact) and becomes the bottleneck for fourth order (Four-Phonon [?] stores real-space $\Phi^{(4)}$ with a cutoff). The stochastic kernel never stores the tensor but re-derives the contraction at every Lanczos application, with statistical noise correlated between nearby fine wavevectors. The price and the payoff are the two sides of the same coin: this is the only route that survives when $\Phi^{(n)}$ cannot be materialized at all — the design restriction of this report.

Where the cutoff enters. Supercell codes control the range by construction (the cutoff *is* the support); coarse- q codes control it by re-centering, and aliasing is the failure mode of skipping that step. The tensor-free estimator inherits the second failure mode: the plain window is exactly the uncentered transform. One genuine novelty here is the pre-filter of Sec. ??: the stochastic correlations are dressed with mode-diagonal propagators f_ψ , so the object being interpolated only acquires the range of $\Phi^{(3)}$ *after* the f_Γ filter is moved onto the coarse-grid fields. Tensor codes never face this problem because they interpolate the bare tensor.

Sum rules: sequential projections vs. one-shot constraints. Tensor codes apply centering, ASR, and symmetrization as separate, non-commuting projections (hence **Center()** *then* **Apply_ASR()**); **symfc**/ALAMODE build the constraints into the fitted basis. The window design of Sec. ?? imposes the centering target, the three kernel ASRs, the partition of unity, and the pair symmetry simultaneously as linear constraints — the same philosophy as **symfc**, applied

to the image-assignment kernel rather than to the force constants. The price is the projection distance of Sec. ??; the gain is that the commensurate limit and the Γ -point ASR hold to machine precision *configuration by configuration*, with zero statistical noise.

What is genuinely new here. The interpolated object is not a bare force-constant tensor but the resummed TDSCHA estimator built from the SSCHA ensemble — which is why the scaling laws $s_3 = \sqrt{N_c/N_f}$, $s_4 = N_c/N_f$ of Sec. ?? and the pre-filter exist at all. None of the external codes has an analogue, because none of them interpolates a stochastic, resummed quantity. Conversely, none of them — and none of the modes here — treats the long-range polar part of $\Phi^{(3)}$ analytically; that limitation is shared across the whole landscape (Sec. ??).

14.5 Recommendation

Our opinion, ordered by regime:

1. **If the tensor can be materialized, use the tensor.** Build (or extract from the SSCHA ensemble) the real-space $\Phi^{(3)}$, center it with atom-resolved minimal-image geometry, project the ASR, and Fourier-evaluate on the target mesh. This is the method phono3py, ShengBTE, D3Q/thermal2, and `cellconstructor` all converged on; it is deterministic, exactly on-grid, reusable, and its error control is transparent (range/cutoff convergence). On this branch it is the `d3_mode="tensor"` oracle, and it is what we benchmark everything against.
2. **If the tensor cannot be stored, reproduce its Fourier sum on the estimator.** The best tensor-free method is the one whose effective kernel equals the centered, ASR-projected image assignment: for D3 today, `atomic_delta` (SnTe $2^3 \rightarrow 4^3$: 37.52 cm^{-1} vs. oracle 37.65); in the fold/unfold formulation, the atom-centred cardinal kernel (38.04 cm^{-1} stochastic, 37.72 cm^{-1} noise-free, vertex reconstruction 2.9×10^{-5}) is the single-pass, coarse-cost realization of the same principle and the most promising direction, pending its extension beyond the D3 Γ slice. For D4, the four-leg factorized kernel of Sec. ?? is the designed candidate; it remains unimplemented.
3. **Never interpolate vertex samples piecewise-linearly.** Trilinear corner averaging has a rigorous amplitude ceiling for any coupling centred between lattice points (the $\mathbb{Z}_2$ class of Sec. ??), fails on a real material in the wrong direction (46.11 cm^{-1}), and is used by no production code. Piecewise-linear maps are safe only for real, non-negative integrands (tetrahedron integration), never for complex vertices.

The unifying rule is simple: *treat the vertex like the dynamical matrix*. Nobody interpolates $D(\mathbf{q})$ linearly between coarse q points, because the real-space force constants are the compact object; the anharmonic vertex obeys the same logic at rank three and four. Centered Fourier interpolation — through a stored tensor when possible, through a tensor-free kernel when necessary — is the best method; everything else this report tested is either an approximation of it (windows, cardinal kernels) or a failure mode (plain, trilinear).

15 Limitations and outlook

The method is deliberately tensor-free, but it still has clear limits.

1. Long-range polar parts of $\Phi^{(3)}$ are not separated analytically. This is the same limitation as the standard `Spectral.py` d_3 route. LO–TO corrections for the interpolated dynamical matrix are future work.

2. The fine mesh does not create new microscopic information. The anharmonic content is still limited by the coarse supercell; nearby fine wavevectors have correlated stochastic noise. The physical gain is the dense two-phonon continuum and the accurately interpolated harmonic dispersion.
3. The ASR constructions here enforce translational sum rules only, as `Apply_ASR` does. Rotational and Born–Huang invariances are not imposed.
4. Designed kernels assume the relevant anharmonic correlations are resolved inside the coarse Wigner–Seitz cell. If the range reaches the boundary, the correct fallback is the plain estimator or a larger coarse cell.
5. Distributed loading, IR/Raman intensity prefactors for the fine mesh, and off-mesh perturbation wavevectors are engineering extensions. The NUDFT field construction already supports arbitrary wavevectors, so spectral maps along paths are a natural next target.
6. The D4 interpolation for underresolved real-material cells remains the main open implementation item. The report gives the four-leg kernel-factorization route and proves its identities, but the runtime `factor4` kernel has not yet been implemented.

A Derivation of the effective interpolation kernel

We derive Eq. (??) in one dimension (the three-dimensional case is the per-dimension product). Let $u(s)$, $s \in \mathbb{Z}$, be the supercell-periodic configuration, $u(s + L) = u(s)$, and let the windowed field at wavevector $\tilde{q}$ (in units where the phase is $e^{-2\pi i \tilde{q}s}$) be

$$\tilde{x}_W(\tilde{q}) = \frac{1}{\sqrt{L}} \sum_s W(s) e^{-2\pi i \tilde{q}s} u(s), \quad (39)$$

with W supported on one period ($W = 1$ on $[0, L)$ for the plain estimator). Consider the three-field product with windows (z, w, v) on the slots and exact conservation $\tilde{q}_1 + \tilde{q}_2 = \tilde{q}_p$:

$$P = \overline{\tilde{z}(\tilde{q}_p)} \tilde{x}_w(\tilde{q}_1) \tilde{y}_v(\tilde{q}_2) = \frac{1}{L^{3/2}} \sum_{s_1 s_2 s_3} z(s_3) w(s_1) v(s_2) e^{2\pi i (\tilde{q}_p s_3 - \tilde{q}_1 s_1 - \tilde{q}_2 s_2)} \bar{u}(s_3) u(s_1) \delta f(s_2). \quad (40)$$

Taking the ensemble average and using translation invariance, $\langle \bar{u}(s_3) u(s_1) \delta f(s_2) \rangle = c_3(s_1 - s_3, s_2 - s_3)$ with c_3 evaluated at the *actual* integer differences (the physical correlation of the periodic system, which itself is the image sum $c_3(\delta_1, \delta_2) = \sum_{n_1 n_2} c_3^\infty(\delta_1 + n_1 L, \delta_2 + n_2 L)$ of the bulk correlation). Substituting $\delta_i = s_i - s_3$ and using $\tilde{q}_p - \tilde{q}_1 - \tilde{q}_2 = 0$ *exactly*, the s_3 dependence of the phase cancels:

$$\langle P \rangle = \frac{1}{L^{3/2}} \sum_{\delta_1 \delta_2} \underbrace{\left[\sum_{s_3} z(s_3) w(s_3 + \delta_1) v(s_3 + \delta_2) \right]}_{S_{z,w,v}(\delta_1, \delta_2)} c_3(\delta_1, \delta_2) e^{-2\pi i (\tilde{q}_1 \delta_1 + \tilde{q}_2 \delta_2)}. \quad (41)$$

Because the windows are zero-padded (support $\leq L$), the differences δ_i range over $(-L, L)$ and each *physical* image of c_3^∞ is weighted by S at its actual (unwrapped) difference and carries its actual phase — this is where the centering choice lives. Two consequences:

- **Commensurate limit.** If $\tilde{q}_i$ are commensurate, all images of a difference class $(\delta_1, \delta_2) \bmod L$ share the same phase, so only the class sums $\sum_{\mathbf{n}} S(\delta + L\mathbf{n})$ matter. If they equal L (partition of unity), Eq. (??) collapses to the exact coarse average — for *any* window design satisfying the constraint. The plain window satisfies it identically ($S = L - \text{spread}$, class sums $= L$).

- **Off-grid.** The kernel S/L is the image-assignment rule of the interpolation. The minimal-image target of Sec. ?? makes it the indicator of the most compact image (ties split), reproducing zero-padded tensor centering.

The same algebra applies verbatim to every D3 term (the y field can sit in any slot) and, with one more field, to the D4 terms, whose kernel is the window quadruple correlation. Note also that no origin appears in Eq. (??): translating the window (equivalently, rolling the data) leaves the expectation invariant, so origin averaging only reduces variance.

B Gaussian integration by parts and the pre-filter identity

For the Gaussian density ρ_Υ with covariance $\langle u_a u_b \rangle = \Upsilon_{ab}^{-1}$, Wick/Isserlis integration by parts gives, for any observable $F(u)$, $\langle u_a F \rangle = \Upsilon_{ab}^{-1} \langle \partial F / \partial u_b \rangle$. Applying it twice to the D3 average (with $\delta f = f - f_{\text{SSCHA}} - \langle f \rangle$, whose second derivative equals that of the full force $f = -\partial V / \partial R$):

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{ab}^{-1} \underbrace{\langle \delta f_c \rangle}_{=0} + \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \left\langle \frac{\partial^2 \delta f_c}{\partial u_{a'} \partial u_{b'}} \right\rangle = -\Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial_{a'b'c}^3 V \rangle. \quad (42)$$

In the SSCHA mode basis Υ^{-1} is diagonal with eigenvalues $f_\psi(\mathbf{q}, \nu) = (1 + 2n_{\mathbf{q}\nu})/2\omega_{\mathbf{q}\nu}$: every *displacement* leg of the correlation carries a propagator factor f_ψ , while force legs carry none. Since the Lanczos kernel multiplies every displacement leg by $f_\Upsilon = 1/f_\psi$ (the $r_1 = f_\Upsilon x$ fields), the combination it averages is the *bare* vertex contraction. The pre-filter of Sec. ?? simply moves the f_Υ multiplication *before* the off-grid transform: on the coarse grid $f_\Upsilon x$ is computed exactly (mode-diagonal), and its real-space transform is a configuration whose correlations with y are contractions of the short-ranged $\Phi^{(3)}$ (in the Cartesian basis, before any polarization projection — the projections are applied at the fine wavevectors analytically). Off-grid, interpolating the filtered fields therefore interpolates a Φ -ranged object; the exact fine-side factors $f_\psi(\tilde{\mathbf{q}})$ that the kernel formulas still require are supplied analytically (folded into the $\alpha^{(1)}$ blocks and set to unity in the kernel tables).

For the four-field averages the same expansion produces, besides the connected $\langle \partial^4 V \rangle$ term, disconnected pieces proportional to $\langle u \otimes \delta f \rangle = \Upsilon^{-1} (\langle \partial^2 V \rangle - \Phi)$, which vanish when Φ is the converged SSCHA matrix. This is the stationarity condition under which the uniform vertex rescaling of Sec. ?? is exact.

C Reproducibility

All numbers and figures in this report are produced by the scripts in `tests/test_interpolation/` (toy model and physics tests) and the benchmark/figure scripts referenced in the repository (`report/interpolation/scripts/`, including `asr_data.py` and `asr_figs.py` for Sec. ??); the decisive physics comparisons are enforced as regression tests (`test_physics_interp.py`, `test_windows.py`, `test_asr_windows.py`). The atom-resolved geometry checks are in `test_atomic_windows.py`. The SnTe oracle and response numbers of Sec. ?? are tracked in `SnTe_FF/Spectral/TDSCHA_Interpolate/issue`; the relevant scripts/data are `kernel_study.py`, `atomic_d3_N300_s20.dat` and `atomic_d3_N1000_s30.dat`. The atom-centred Fourier audit of Sec. ?? is reproduced by `scripts/snte_atom_centered_fourier.py` and `scripts/bench_snte_atom_fourier.py`; its structural cardinal, transpose-symmetry, and Hermiticity checks are in `tests/test_trilinear/test_trilinear_structural.py`.

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